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Digital Twin for railway network simulation

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Contents

Chapter 1

Introduction

Railway networks are essential infrastructures for passenger mobility, freight transport, and sustainable transportation. In dense networks such as the Belgian railway system, operating trains efficiently and reliably is a complex task. Many trains, stations, tracks, and operational constraints interact continuously, while disruptions in one part of the network can propagate to other services and affect punctuality at a wider scale.

This complexity is both infrastructural and operational. Trains must follow predefined routes and timetables, share limited track capacity, respect safety constraints, and interact with elements such as switches, platforms, and stations. As a result, railway operators and researchers increasingly rely on digital tools to better understand, simulate, and analyse railway systems.

In this context, the concept of a digital twin has gained increasing attention. Informally, a digital twin can be understood as a virtual counterpart of a physical system, connected to data and used to observe, simulate, and analyse its behaviour. Applied to railways, such a representation may combine infrastructure data, train movements, operational constraints, and traffic conditions. Simulation is therefore a central component of this vision, since it provides a controlled environment in which railway operations can be reproduced, tested, and compared with real data.

This thesis focuses on the construction of a simulation-oriented railway digital twin using SUMO. The objective is not to build a complete real-time digital twin, but to study how heterogeneous railway infrastructure and operational datasets can be transformed into simulation-ready artefacts. The work also investigates how different levels of railway network representation influence the feasibility and realism of the resulting simulation.

1.1 Research Problem and Objectives

Despite the availability of railway data and simulation platforms, constructing a realistic railway simulation from real-world datasets remains difficult. Railway data is heterogeneous by nature: infrastructure datasets may describe stations, tracks, switches, platforms, and geographical geometries, while operational datasets describe train movements, schedules, arrival and departure times, and punctuality records. These datasets do not always share the same structure, granularity, or level of detail. Integrating them into a single simulation framework therefore requires several processing, modelling, and approximation steps.

A first challenge concerns the representation of the railway infrastructure. At a high level, the network can be represented as a graph in which stations are nodes and railway links are edges. This abstraction is useful because it simplifies route reconstruction and supports large-scale simulation. However, it does not represent detailed infrastructure elements such as platforms, switches, parallel tracks, or station-level routing constraints.

A second challenge appears when moving toward a more detailed representation. At the micro level, tracks, platforms, switches, and routing structures must be represented explicitly. This provides a more realistic description of the infrastructure, but it also requires a topologically consistent network. Platforms must be correctly associated with tracks, and valid paths must exist between consecutive stations. Even small inconsistencies in the reconstructed infrastructure can prevent trains from being routed correctly.

A third challenge concerns the integration of real operational data. Punctuality data provides information about train stops, planned and observed times, and delays, but it does not directly provide the exact physical route followed by each train. Train journeys must therefore be reconstructed from station sequences and mapped onto the generated simulation network. This process is relatively robust at the macro level, but becomes more fragile at the micro level because each movement must correspond to a valid sequence of physical track segments.

These challenges are addressed through the following research sub-questions:

RQ1. How can heterogeneous railway infrastructure and operational datasets be transformed into SUMO-compatible simulation components?

RQ2. To what extent can a macro-level station-to-station representation support the integration of real operational data into railway simulation?

RQ3. To what extent can a micro-level infrastructure representation support detailed railway simulation based on tracks, platforms, switches, and routing structures?

RQ4. What are the main limitations and failure points encountered when integrating real operational data into micro-level railway simulation?

The main objective of this thesis is to design and implement a data-driven pipeline for constructing a simulation-oriented railway digital twin. This pipeline is evaluated through two complementary modelling approaches. The macro-level model represents the railway network as a station-to-station graph and is used to validate the integration of real operational data into simulation. The micro-level model represents the infrastructure in greater detail and is used to study the feasibility of more realistic railway simulation.

The expected outcome is therefore not a perfect reproduction of the Belgian railway network, but a structured and evaluated prototype showing how railway data can be transformed into SUMO-compatible simulation models. The contribution lies in the construction of the modelling pipeline, the comparison between macro-level and micro-level representations, and the identification of the technical challenges that must be addressed to progress toward more complete railway digital twins.

1.2 Thesis Structure

The remainder of this thesis is organized as follows.

Chapter 2 presents the state of the art. It introduces the concept of digital twins, first from a general perspective and then in the context of railway systems. It also reviews the main components of railway infrastructure and operations, discusses common approaches for railway network representation, and presents existing railway simulation tools, with particular attention to SUMO.

Chapter 3 describes the proposed methodology. It positions the work with respect to the research gaps identified in the literature, presents the overall framework of the approach, justifies the choice of SUMO, and describes the infrastructure and operational datasets used in the implementation.

Chapter 4 details the implementation of the proposed approach. It is divided into two main parts. The first presents the macro-level model, including data processing, network generation, trip generation, and simulation. The second presents the micro-level model, including infrastructure modelling, switch detection, platform detection, track segmentation, dual graph construction, trip generation, and simulation.

Chapter 5 presents the results obtained from the implemented models. It analyses dataset coverage, infrastructure modelling quality, simulation results, and computational performance. Particular attention is given to the difference between successful macro-level integration and the difficulties encountered when using real operational data at the micro level.

Chapter 6 discusses the main limitations of the proposed methodology. It provides a more detailed analysis of issues related to data quality, modelling assumptions, route reconstruction, SUMO constraints, and scalability.

Chapter 7 concludes the thesis by summarizing the main contributions and findings. It also discusses the practical implications of the work and presents future research directions, including counterfactual simulation, track work planning, real-time forecasting, incident propagation, and framework generalization.

Chapter 2

State of the Art

2.1 Digital Twins

Digital twins have become one of the central concepts associated with the digitalization of complex industrial systems ([10]). The term is used in many domains, including manufacturing, aerospace, healthcare, smart cities, energy systems, and transportation ([10, 26, 24]). Although the exact definition may vary depending on the application domain, the general idea remains the same: a digital twin is a digital representation of a physical system that is connected to data and used to analyse, simulate, monitor, or support decisions about the real system ([15, 10]).

In the context of this thesis, the concept of a digital twin is particularly relevant because railway networks are complex, dynamic, and data-rich systems. They involve physical infrastructure, moving vehicles, operational rules, timetables, delays, maintenance constraints, and safety requirements. A digital twin can therefore provide a structured framework to combine infrastructure data, operational data, simulation models, and analytical tools into a coherent representation of the railway system.

2.1.1 Definition, Capabilities and Limitations

The concept of digital twin was initially introduced in the context of product lifecycle management and later expanded to other engineering and industrial domains. A commonly accepted definition describes a digital twin as a virtual representation of a physical object, process, or system that is connected to its physical counterpart through data exchange and that can be used throughout the lifecycle of the system. This definition emphasizes that a digital twin is not merely a static model or a visualization tool. Its distinguishing characteristic is the relationship between the physical entity and its digital counterpart.

A digital twin can therefore be understood as the combination of three fundamental components: the physical system, the virtual system, and the connection between them. The physical system corresponds to the real-world object or process being represented. The virtual system is the digital model that describes the structure, behaviour, and state of the physical system. The connection corresponds to the data flow that links both entities. This connection may be unidirectional, when the physical system only updates the virtual model, or bidirectional, when the digital model can also influence decisions or actions applied to the physical system ([25]).

A more formal way to describe a digital twin is to represent it as a system composed

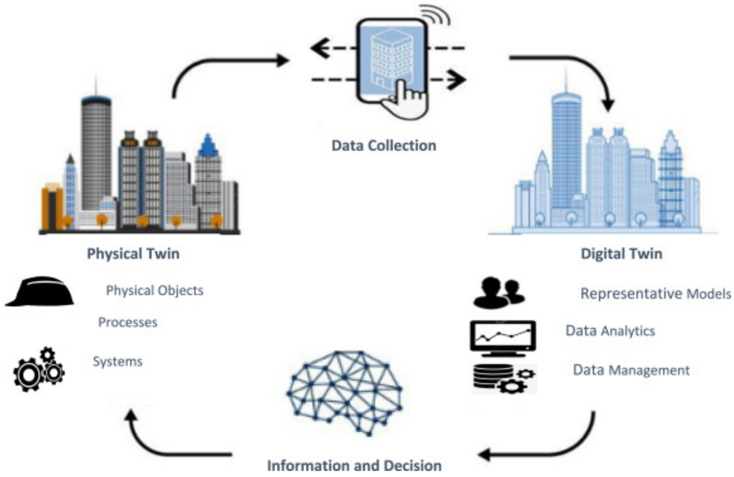

Figure 2.1: General architecture of a digital twin.

of several interacting elements ([32, 10]):

$$DT = (P, V, D, C, S) \tag{2.1}$$

where P denotes the physical system, V the virtual representation, D the data associated with the system, C the connection mechanisms between the physical and virtual entities, and S the services built on top of the twin, such as simulation, prediction, visualization, diagnosis, or optimization. This representation highlights that the digital twin is not only a model, but an architecture in which data, models, and services are combined to support decision-making.

This distinction is important because the term digital twin is sometimes used broadly to describe any digital model of a real system. However, several authors distinguish between a digital model, a digital shadow, and a digital twin ([21]). A digital model is a digital representation of a system without automatic data exchange with the physical system. A digital shadow receives data from the physical system, but the information flow is mainly one-way. A digital twin represents a more advanced form, where the digital representation is continuously or regularly updated from the physical system and can support feedback, analysis, or operational decisions ([21]). In this thesis, the proposed work is closer to a simulation-oriented digital twin foundation: it focuses on constructing a coherent virtual railway representation from real infrastructure and operational data, which is a necessary step before reaching a fully real-time and bidirectional digital twin.

Digital twins are usually associated with several capabilities. The first is monitoring, which consists of observing the state of a system using data collected from sensors, logs, operational records, or information systems. The second is simulation, where the digital representation is used to reproduce the behaviour of the physical system under controlled conditions ([10, 30]). The third is prediction, where historical and real-time data are used to estimate future states, failures, delays, or performance indicators ([10, 26, 24]). The fourth is optimization, where alternative scenarios are evaluated in order to support better decisions. Finally, more advanced digital twins may support control or actuation, where decisions derived from the digital model influence the operation of the physical system.

These capabilities make digital twins particularly attractive for complex systems, but they also introduce significant limitations. A digital twin is only as reliable as the data, models, assumptions, and synchronization mechanisms on which it is based. If the data is incomplete, outdated, noisy, or inconsistent, the digital twin may provide misleading results. Similarly, if the model is too simplified, it may fail to reproduce important physical or operational constraints. On the other hand, if the model is too detailed, it may become computationally expensive and difficult to maintain.

Another limitation concerns validation. Since a digital twin is expected to represent a real system, it must be evaluated against real observations. This is challenging because real systems are often affected by hidden variables, incomplete measurements, operational exceptions, and human decisions that are not always captured in the available data. For this reason, the development of a digital twin is usually incremental. A first version may focus on static representation and simulation, while later versions progressively integrate real-time data, predictive models, and optimization services.

2.1.2 Recent Developments and Applications

The development of digital twins has accelerated with the progress of several enabling technologies. The IoTs allows physical systems to generate continuous streams of data ([10]). Cloud computing and distributed architectures provide the infrastructure required to store and process large volumes of information. Artificial intelligence and machine learning make it possible to extract patterns, predict future states, and detect anomalies. Simulation tools provide an environment in which alternative scenarios can be tested without disrupting the real system. Finally, visualization technologies, including dashboards, 3D interfaces, and geographic information systems, help users interpret the state and behaviour of the DT.

Digital twins have first been widely adopted in manufacturing, where they support product design, production monitoring, maintenance planning, and process optimization. In smart manufacturing, a digital twin can represent a machine, a production line, or an entire factory. It can be used to monitor equipment status, simulate production scenarios, predict failures, and improve resource allocation ([30]). This domain is particularly important in the literature because it provides many of the conceptual foundations later reused in other fields.

In aerospace and mechanical engineering, digital twins are used to monitor the condition of critical assets and support predictive maintenance. Aircraft, engines, turbines, and other complex mechanical systems generate large quantities of operational data during their lifecycle. A digital twin can integrate this data with physics-based or data-driven models in order to estimate wear, detect abnormal behaviour, and plan maintenance interventions before failures occur. Similar principles are found in energy systems, where digital twins are used for turbines, power plants, and renewable energy infrastructure ([24]).

Digital twins are also increasingly studied in healthcare and smart cities. In healthcare, the objective may be to represent medical devices, hospital workflows, or even patient-specific physiological models. In smart cities, digital twins can integrate mobility data, infrastructure data, environmental data, and urban services to support planning and decision-making. These examples show that digital twins are not limited to individual physical objects. They can also represent complex systems composed of many interacting components.

Across these application domains, digital twins generally serve four main purposes. First, they improve visibility by providing a structured representation of a system that may otherwise be difficult to observe directly. Second, they support experimentation by allowing users to test what-if scenarios in a virtual environment. Third, they improve anticipation by enabling predictive analysis based on past and current data. Fourth, they support decision-making by comparing alternative actions before they are applied to the real system.

However, the maturity of digital twin implementations varies significantly. In some cases, the term digital twin refers to a highly connected and dynamic system with real-time data exchange. In other cases, it refers to a simulation model or a dashboard that is only periodically updated. This variation shows that the digital twin concept should not be treated as a single technology, but rather as a framework that can be implemented at different levels of maturity. For this thesis, the focus is placed on the simulation and data integration layers, which are essential prerequisites for more advanced digital twin capabilities such as real-time forecasting, optimization, and decision support.

2.1.3 Applications in Railway networks

Railway systems are particularly suitable candidates for digital twin approaches. They combine physical infrastructure, operational processes, moving assets, strict safety constraints, and large volumes of data. A railway network includes tracks, switches, signals, stations, platforms, rolling stock, timetables, passenger flows, maintenance operations, and traffic management rules. These elements interact continuously, and a local disturbance can propagate through the network. As a result, railway operators require tools that can represent the current state of the network, simulate traffic behaviour, evaluate disruptions, and support planning decisions.

Digital twins for railway networks can be applied at different levels of granularity. At the asset level, a digital twin can represent individual components such as tracks, switches, bridges, tunnels, trains, or signalling equipment. This type of twin is particularly useful for condition monitoring and predictive maintenance. Sensor data, inspection records, and maintenance logs can be used to estimate the health of the asset and identify when intervention is required.

At the infrastructure level, a digital twin can represent the topology of the railway network. This includes the spatial structure of tracks, stations, platforms, junctions, and operational segments. Such a representation is necessary to analyse accessibility, capacity, routing possibilities, and infrastructure conflicts. In this thesis, this level is central because the objective is to transform real railway infrastructure data into simulation-ready network models.

At the operational level, a railway digital twin can represent train movements, schedules, delays, and traffic interactions. This makes it possible to analyse how trains circulate through the network, how delays propagate, and how infrastructure constraints affect service quality. A simulation-based digital twin can also be used to evaluate what-if scenarios, such as modified timetables, temporary track closures, traffic disruptions, or infrastructure works.

Several applications can be identified in the railway domain. The first is predictive maintenance. Railway infrastructure is expensive to inspect and maintain, and failures can have significant consequences on safety and service reliability. A digital twin can combine sensor measurements, historical degradation data, and maintenance records to anticipate failures and optimize maintenance planning. The second application is traffic management. By reproducing train movements in a virtual environment, a digital twin can help evaluate congestion, detect bottlenecks, and test operational strategies. The third application is delay analysis and propagation modelling. Since railway operations are strongly interconnected, a delay affecting one train can influence many others. A digital twin can help identify where delays originate and how they spread across the network.

A fourth application is scenario testing. Railway operators often need to evaluate changes before implementing them in the real network. Examples include timetable modifications, station layout changes, temporary closures, track works, or emergency response strategies. A simulation-based digital twin makes it possible to test such scenarios in a controlled environment and compare their expected impact. This capability is especially relevant for this thesis, since the proposed approach focuses on generating railway simulations from real infrastructure and operational data.

Railway digital twins also raise specific challenges. Railway networks are safety-critical systems, which means that simplified models must be used with care. Accurate representation of tracks, switches, signals, platforms, and routing constraints is necessary when the objective is to reproduce detailed operational behaviour. Data availability is another major challenge. Infrastructure data may be incomplete, private, heterogeneous, or difficult to align with operational data. Finally, railway systems operate at different spatial and temporal scales. A digital twin may need to represent long-distance network connectivity, station-level infrastructure, and real-time traffic dynamics at the same time.

For these reasons, railway digital twins often require a layered approach. A macro-level representation can be used to model station-to-station connectivity and global train movements. A micro-level representation can then be used to represent detailed infrastructure such as tracks, switches, and platforms. This distinction is directly reflected in the methodology of this thesis, where both macro-level and micro-level railway models are developed. The macro-level model provides a simplified and scalable representation of the network, while the micro-level model aims to capture more detailed infrastructure constraints. Together, these two levels contribute to the construction of a simulation-oriented foundation for a railway digital twin.

In summary, digital twins provide a relevant conceptual and technical framework for railway simulation. They offer a way to combine infrastructure data, operational data, simulation models, and analytical tools into a unified system. However, building a complete railway digital twin requires several intermediate steps: reconstructing the infrastructure, generating valid train routes, integrating real operational data, validating simulation outputs, and identifying the limitations of the model. The work presented in this thesis focuses on these foundational steps, with the objective of preparing the ground for future digital twin applications such as real-time forecasting, disruption analysis, and decision support.

2.2 Railway Systems

Railway systems are complex socio-technical systems composed of physical infrastructure, rolling stock, operational rules, control systems, and data-driven decision processes. Unlike road traffic, where vehicles usually have a high degree of freedom in their trajectories, trains are constrained to follow predefined tracks, respect signalling rules, and operate according to planned timetables. This makes railway systems highly structured, but also highly constrained.

Understanding railway systems is essential for the construction of a railway digital twin. A digital twin does not only require a geometric or topological description of the network, it must also account for how trains move, how infrastructure is used, how conflicts are avoided, and how delays propagate.

2.2.1 Railway infrastructure

Railway infrastructure refers to the physical and technical components that allow trains to circulate safely and efficiently. It includes tracks, switches, junctions, stations, platforms, signalling systems, electrification equipment, and communication systems. From a modelling perspective, infrastructure defines the spatial structure of the network and the constraints under which train movements can occur.

In railway simulation, the level of detail used to represent infrastructure depends on the objective of the model. A high-level model may only represent stations and the connections between them, which is sufficient for analysing large-scale connectivity or approximate train movements. A more detailed model must represent tracks, switches, platforms, signals, and route conflicts, which is necessary for studying station-level operations, capacity, and realistic routing constraints.

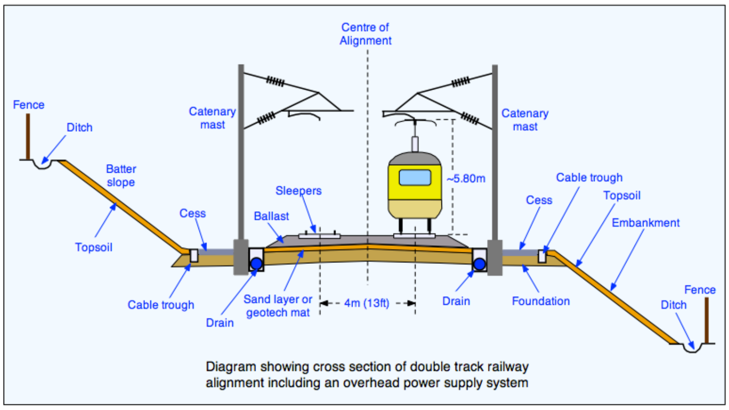

Figure 2.2: Schematic representation of railway infrastructure components.

Tracks

Tracks are the fundamental physical component of a railway network. They guide the movement of trains and define the possible spatial paths through the infrastructure. A railway track generally consists of rails, sleepers, ballast or slab track, and the supporting formation. From an operational point of view, a track is not only a physical object but also a capacity resource: only a limited number of trains can use the same section of track within a given time interval.

In modelling terms, tracks can be represented at different levels of abstraction. At a macroscopic level, a track may be abstracted as a connection between two stations. This representation is useful when the objective is to model station-to-station connectivity or analyse large-scale train movements. At a microscopic level, however, tracks must be represented as individual segments with geometry, direction, length, speed constraints, and connections to neighbouring track elements. This level of detail is required when the objective is to compute feasible routes through complex station areas or to represent switches and platform access.

Track directionality is also important. Some railway lines are designed for traffic in one direction only, while others allow bidirectional operation. Even when tracks are physically bidirectional, signalling and operational rules may restrict their use. In simulation, this distinction affects route generation because a path that is geometrically possible may not be operationally valid. Therefore, a railway model must distinguish between physical connectivity and usable operational connectivity.

Another important aspect is the distinction between plain line tracks and station or yard tracks. Plain line tracks connect railway nodes over longer distances and are usually associated with running movements. Station tracks, sidings, and yard tracks are often used for stopping, overtaking, shunting, or storing rolling stock. Their operational behaviour differs from mainline tracks and may require additional modelling assumptions.

For this thesis, the concept of track is central in the micro-level model. Individual track segments form the basis of the reconstructed infrastructure graph. Their connections determine whether a train path can be computed between two stations or platforms. In contrast, the macro-level model abstracts tracks into station-to-station links, which reduces complexity but omits detailed infrastructure constraints.

Switches and Junctions

Switches, also called points or turnouts, are infrastructure elements that allow trains to move from one track to another. They are essential for routing because they introduce branching possibilities in the network. Without switches, a railway network would be a set of independent lines with limited operational flexibility. Switches enable trains to enter stations, change tracks, access sidings, overtake other trains, or follow alternative routes.

A switch is usually composed of movable rails that guide the wheels of a train toward one of several possible paths. From a modelling perspective, a switch can be understood as a local decision point in the infrastructure graph. It connects an incoming track to two or more outgoing tracks, depending on its configuration. Junctions are larger structures composed of several switches and crossings, often located near stations, yards, or intersections between railway lines.

[Figure 2.4 should be inserted here: Example of a railway switch/turnout showing the straight route and diverging route.]

Switches introduce important operational constraints. A train can only pass through a switch if it is correctly aligned for the intended route. Furthermore, two trains cannot use conflicting routes through the same junction at the same time. These constraints are usually enforced by signalling and interlocking systems. In detailed railway simulations, switches and junctions must therefore be modelled not only as graph connections, but also as shared resources that may generate conflicts.

The correct identification of switches is particularly important for digital twin construction. If switches are missing from the infrastructure model, some valid routes may not be found. Conversely, if false switches are detected, the model may allow unrealistic routes. This is one of the reasons why micro-level railway modelling is more challenging than macro-level modelling. It requires detecting and representing the internal structure of the network with sufficient accuracy.

In this thesis, switches play an important role in the micro-level infrastructure model. They are detected from the topology of the railway network and converted into simulation-compatible nodes. This allows the generated network to represent routing possibilities at a finer level than the macro-level model.

Figure 2.3: Example of a railway switch/turnout showing the straight route and diverging route.

Stations and Platforms

Stations are operational nodes where trains may stop, passengers may board or alight, and services may be regulated. They are often among the most complex elements of a railway network because they combine infrastructure, passenger flows, operational constraints, and timetable requirements. A station may contain several tracks, multiple platforms, switches, signals, and access routes.

In a macroscopic representation, a station can be modelled as a single node. This is useful when the objective is to analyse connections between stations or reconstruct train journeys from operational data. However, such a representation hides the internal structure of the station. It does not indicate which platform a train uses, whether two trains can stop simultaneously, or whether the access routes to different platforms conflict.

In a microscopic representation, a station must be decomposed into tracks, platforms, stopping positions, and access routes. Platforms define where passenger exchanges occur. They are important for simulation because a train stop must be attached to a specific track or lane in the simulation network. If a platform is incorrectly associated with a track, the simulated train may stop at the wrong location or may not be able to reach the stop at all.

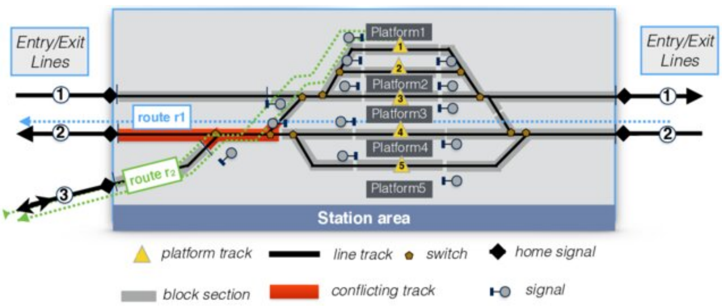

Figure 2.4: Example of a station layout

Stations also influence capacity. Dwell time, platform availability, route conflicts at station throats, and the number of available tracks determine how many trains can be handled within a given period. Large stations may become bottlenecks even if the lines around them have sufficient capacity. For this reason, station modelling is a key component of realistic railway simulation.

In this thesis, stations are used differently in the macro-level and micro-level approaches. In the macro-level model, stations are represented as nodes connected by edges. In the micro-level model, stations are associated with platforms and track segments. This distinction makes it possible to compare a simplified connectivity-based representation with a more detailed infrastructure-based representation.

Signaling Systems

Signalling systems are used to ensure the safe movement of trains through the railway network. Their main role is to prevent conflicting train movements and to maintain safe separation between trains. Signalling systems communicate movement authorities, speed restrictions, route availability, and stopping instructions to train drivers or onboard systems.

Traditional signalling systems divide the railway line into block sections. Under normal operation, only one train is allowed to occupy a block section at a time. A train may proceed only if the section ahead is clear and if the route has been correctly set. This principle prevents rear-end collisions and conflicting movements. In more advanced systems, movement authority may be transmitted directly to the train through cab signalling or automatic train protection systems.

In Europe, signalling and train control are strongly influenced by interoperability requirements. The European Rail Traffic Management System (ERTMS) and the European Train Control System (ETCS) aim to standardize train control and signalling across countries. Such systems are relevant for digital twin research because they generate and rely on structured operational data and because they influence how train movements can be represented.

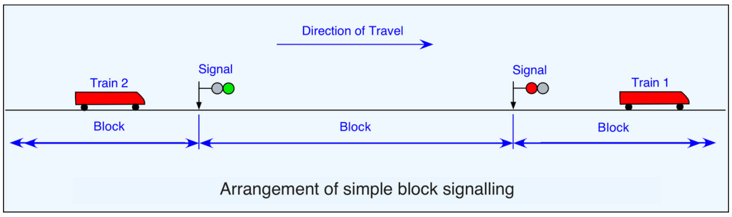

Figure 2.5: Simplified block signalling principle

From a simulation perspective, signalling can be represented at different levels of detail. A simple simulation may ignore signalling and only check whether routes exist. A more realistic simulation must represent block occupation, signal states, route reservations, interlocking constraints, and train separation. The level of signalling detail has a direct impact on the realism of simulated train movements and capacity analysis.

In this thesis, signalling is not fully modelled as an operational control system. However, it remains an important limitation and a key concept for interpreting the realism of the generated simulations. The absence or simplification of signalling means that the simulation can validate connectivity and route generation, but cannot fully reproduce all safety-critical operational constraints of a real railway network.

2.2.2 Railway Operations

Railway operations describe how trains are planned, dispatched, controlled, and managed over the infrastructure. While infrastructure defines what is physically possible, operations define what is actually allowed and scheduled. Railway operations combine timetables, train paths, rolling stock constraints, crew constraints, safety rules, capacity management, and real-time traffic regulation.

For a railway digital twin, operations are essential because the objective is not only to represent the network, but also to reproduce or analyse train movements. Operational data provides information about scheduled and actual train movements, delays, stops, and service patterns. This data can be used to generate simulation scenarios, validate the model, and study the behaviour of the railway system.

Train Movements

A train movement is the circulation of a train from one point of the network to another. It is defined by an origin, a destination, a route, a departure time, intermediate stops, and operational constraints. In railway operations, a train cannot freely choose its path at any time. Its movement must be compatible with the infrastructure, the timetable, signalling rules, and other train movements.

Train movements can be classified according to their operational purpose. Passenger trains usually follow fixed routes and timetables, stopping at stations to allow boarding and alighting. Freight trains may have more variable paths and may be planned according to capacity availability, terminal access, and logistic constraints. Empty stock movements, maintenance trains, and shunting movements introduce additional complexity because they may not correspond directly to passenger service patterns.

In simulation, train movements must be converted into routes and departure times. A route is a sequence of infrastructure elements that a train follows. At a macro level, this may be a sequence of stations. At a micro level, it must be converted into a sequence of track segments or edges. This conversion is one of the central challenges of railway digital twin construction, because real operational records often describe train movements at the station level, while the simulator requires detailed network paths.

The feasibility of a train movement depends on connectivity and timing. Connectivity means that a valid path must exist between consecutive locations. Timing means that the train must be inserted into the simulation at the correct time and must respect operational constraints such as dwell times and travel times. If either the connectivity or timing is inconsistent, the simulation may fail or produce unrealistic behaviour.

Timetables

A railway timetable defines the planned movements of trains over the network. It specifies departure and arrival times, stopping patterns, running times, dwell times, and sometimes platform assignments or path allocations. Timetabling is a central process in railway operations because it organizes the use of limited infrastructure capacity.

Railway timetabling is a constrained optimization problem. A feasible timetable must satisfy commercial objectives, such as service frequency and passenger demand, while also respecting technical constraints, such as minimum headways, track capacity, station capacity, rolling stock availability, and safety margins. In dense networks, small changes to a timetable may affect many other services because trains share infrastructure resources.

Periodic timetables are common in passenger railway systems. In such timetables, services repeat at regular intervals, for example every 30 or 60 minutes. This makes the timetable easier to understand for passengers and easier to coordinate across lines. However, periodicity also introduces additional constraints because connections, crossings, and resource usage must remain consistent across repeated time cycles.

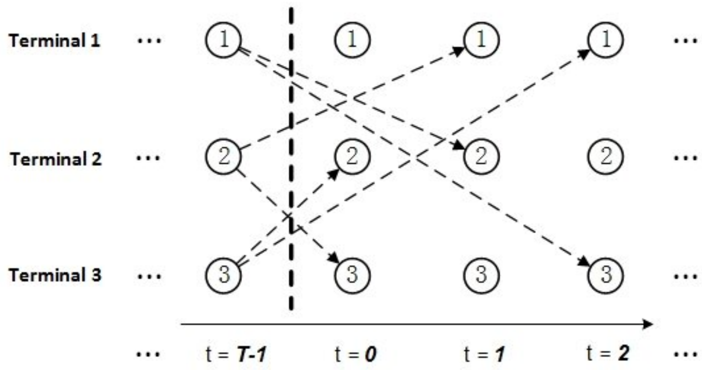

Figure 2.6: Example of a time-space diagram.

A common way to visualize train movements and timetables is the time-space diagram. In such a diagram, one axis represents time and the other represents position along a railway line. Each train is represented by a line. The slope of the line reflects its speed, stops appear as horizontal segments, and crossings or overtakes can be analysed visually. Time-space diagrams are particularly useful for understanding capacity, conflicts, delays, and recovery margins.

Capacity Management

Railway capacity refers to the ability of infrastructure to accommodate train movements within a given period while respecting safety, operational, and quality constraints. Unlike road capacity, railway capacity is highly dependent on timetable structure, train heterogeneity, signalling, station dwell times, and route conflicts. Therefore, capacity is not only a property of the physical infrastructure but also of the way it is used.

A railway line with homogeneous trains running at similar speeds and stopping patterns can usually accommodate more trains than a line where fast, slow, passenger, and freight trains are mixed. This is because heterogeneous traffic increases the need for headways, overtaking, and timetable margins. Similarly, a station with limited platforms or conflicting access routes may reduce capacity even if the surrounding tracks are not saturated.

The UIC 406 method is often used as a reference approach for railway capacity analysis. It evaluates capacity consumption by compressing train paths in a timetable and analysing how much time is occupied by train movements. This approach highlights that capacity is related not only to the number of trains, but also to their ordering, spacing, speed differences, and operational margins.

Capacity management is important for simulation because a simulation may generate routes that are technically connected but operationally unrealistic if capacity constraints are ignored. For example, multiple trains may be assigned to the same track or platform at overlapping times. A detailed simulator should therefore account for infrastructure occupation, route conflicts, and safe separation.

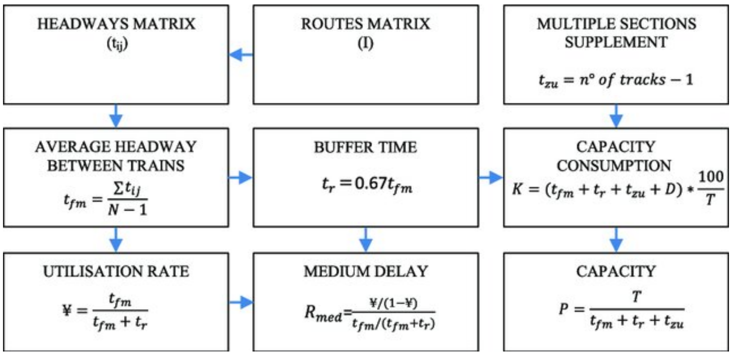

Figure 2.7: UIC 406 method procedure.

In this thesis, capacity is not the main optimization objective, but it remains a relevant concept. The macro-level model provides a simplified view of train circulation, while the micro-level model makes it possible to represent more detailed infrastructure constraints. Future extensions of the work could use the generated simulation framework to study capacity consumption, bottlenecks, and infrastructure works planning.

Delay Propagation

Delay propagation is one of the most important phenomena in railway operations. A primary delay occurs when a train is delayed by an external or local cause, such as technical failure, passenger boarding time, weather, infrastructure problems, or operational incidents. A secondary delay occurs when this initial delay affects other trains through shared infrastructure, rolling stock circulation, crew dependencies, passenger connections, or timetable conflicts.

Railway networks are particularly sensitive to delay propagation because many trains share the same tracks, stations, junctions, and timetable margins. A delayed train may occupy a track section later than planned, preventing another train from using it. It may also miss a planned crossing, delay a connection, or arrive late at a station where the rolling stock is needed for another service. As a result, a local disturbance can spread through the network and affect trains that were not directly involved in the initial incident.

Delay propagation can be studied at different levels. Microscopic approaches simulate train movements and conflicts in detail. Macroscopic approaches analyse delays at the level of stations, lines, or aggregated network regions. Data-driven approaches use historical operational data to estimate how delays evolve over time and space. Network-based approaches represent trains, stations, or events as nodes and study how delays spread through their interactions.

Capacity management is important for simulation because a simulation may generate routes that are technically connected but operationally unrealistic if capacity constraints are ignored. For example, multiple trains may be assigned to the same track or platform at overlapping times. A detailed simulator should therefore account for infrastructure occupation, route conflicts, and safe separation.

In the current work, delay propagation is not fully modelled as a predictive process. However, the reconstruction of infrastructure and train movements is a prerequisite for such applications. Without a valid simulation network and feasible train routes, it is not possible to study how operational disturbances propagate across the railway system.

2.2.3 Railway Network Representation

A railway network can be represented in several ways depending on the modelling objective. The choice of representation determines which phenomena can be analysed and which assumptions are made. For digital twin and simulation purposes, three broad families of representations are particularly relevant: graph-based modelling, physics-based modelling, and logic-based modelling.

These representations are not mutually exclusive. A detailed railway simulator may combine graph-based infrastructure, physics-based train dynamics, and logic-based operational constraints. However, each representation emphasizes a different aspect of the railway system.

Graph-Based Modeling

In this representation, the network is modeled as a directed graph, where nodes represent stations, switches or significant waypoints, and edges correspond to track segments or connections between these nodes.

One the key strengths of graph-based models lies in their compatibility with wellestablished algorithms from graph theory and operation research. These models are particularly effective in macroscopic task planning, such as timetable design, resource allocation, or delay propagation studies, where the physical movement of trains is not needed.

Still, this approach lacks temporal and dynamic granularity. Trains usually modeled as moving instantaneously along edges without continuous time progression constraints. Moreover, aspects like acceleration, braking or signaling rules are often ignored or oversimplified. As a result, they are not effective in applications requiring realistic simulations of train dynamics of precise safety control.

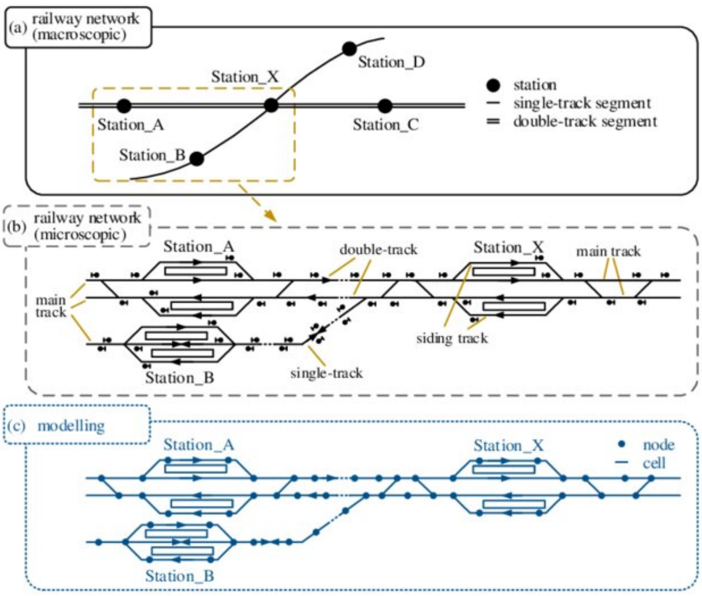

Figure 2.8: Example of a railway station modelling at macro and micro levels.

Physics-Based Modeling

Physics-based models aim to represent the physical behavior of trains and their interactions with the infrastructure. These models simulate train motion as a function of traction force, resistance, braking and inertia.

The strength of this approach is its realistic depiction of train dynamics, which is crucial for applications like energy-efficient timetabling or validation of rolling stock performance under specific track conditions.

Yet, physics-based models ten to focus on individual trains or small-scale simulations due to their computational demands. They do not capture the logical control elements such as interlocking, and they require detailed data like train mass, motor characteristics or speed restrictions. Therefore, they are less suitable for modeling network-wide traffic flow or verifying control logic at a system level.

Logic-Based Modeling

Logic-based models are designed to capture the discrete control logic and safety constraints in trains operations. One major example is the modeling of interlocking systems, which ensures that only safe and conflict-free movements are allowed on the network. These models abstract the railway as a set of finite states and transitions, with logical rules defining when a train can move from one segment to the other based of multiple parameters like the sate of switches, signals or blocks. They are usually expressed using formal methods. Tools like UupaAl and NuSMV allow formal verification of safety properties and correctness conditions.

Logic-based models are useful when the goal is the design and validation of signaling systems. They enable exhaustive exploration of all possible states and transitions to identify potential conflicts or unsafe configurations. They are also essential in the development on DTR, where real-time monitoring an predictive analysis require an accurate model of control logic.

However, this approach requires highly detailed rule definitions to be effective. It can also face scalability challenges as the network size increases, especially when faces with concurrency (multiple trains operation simultaneously).

Each of the three modeling captures different but complementary aspects of a railway system. Combining graph abstraction for routing, physics modules for motion and simulation, and formal logic for safety are increasingly explored in the industry to connect these domains and support robust digital representations of railway networks.

2.2.4 Railway Data Sources

Railway modelling and simulation depend strongly on the availability, completeness, and consistency of data. A digital twin requires data describing both the physical infrastructure and the operation of trains. These two categories are often stored in different systems, follow different formats, and use different levels of granularity. Aligning them is one of the main challenges of railway digital twin construction.

Infrastructure data describes what exists in the network. Operational data describes how the network is used. A simulation-oriented digital twin must combine both: infrastructure data is required to build the network, while operational data is required to generate train movements and validate the behaviour of the simulation.

Infrastructure Data

Infrastructure data includes information about tracks, switches, stations, platforms, signals, speed limits, electrification, geometry, and topology. Depending on the source, this data may be available as geospatial data, tabular data, XML files, engineering databases, or standardized railway formats.

OpenStreetMap and OpenRailwayMap are examples of open geospatial sources that can provide railway infrastructure information. They may include track geometry, station locations, railway line types, electrification, gauge, signals, and other attributes. These sources are useful because they are publicly accessible and cover large areas. However, their completeness and accuracy may vary depending on the region and the contributions of the mapping community.

Institutional data sources, such as infrastructure manager databases, are usually more accurate and complete but are often private or restricted. In Belgium, infrastructure data from Infrabel or related railway stakeholders can provide more reliable information about stations, tracks, platforms, and operational assets. However, such data may require preprocessing, cleaning, transformation, and alignment before it can be used for simulation.

Standardized formats are also important in railway data exchange. RailML and related railway modelling standards provide structured ways to describe infrastructure, rolling stock, timetables, and operational data. These formats are useful because they improve interoperability between tools. However, converting raw data into such formats can require significant modelling effort.

In this thesis, infrastructure data is used at two levels. At the macro level, station and connectivity data are used to build a station-to-station network. At the micro level, detailed track and platform data are used to reconstruct a finer infrastructure graph. This distinction reflects the difference between global network connectivity and detailed simulation-ready infrastructure.

Operational Data

Operational data describes the actual or planned movement of trains. It may include timetables, train identifiers, station stops, arrival and departure times, delays, platform assignments, rolling stock information, and disruption records. Operational data is essential for transforming a static railway infrastructure model into a dynamic simulation scenario.

Timetable data provides the planned structure of train services. It defines which trains should run, when they should depart, where they should stop, and how they should connect to other services. Punctuality data provides information about actual train movements and deviations from the planned timetable. It can be used to analyse delay patterns, validate simulation results, or generate realistic train routes.

Operational data is often less straightforward to use than infrastructure data. It may contain missing records, inconsistent station identifiers, cancelled trains, duplicated entries, or timing errors. It may also describe movements at the station level, while the simulation requires detailed paths at the track level. Therefore, data cleaning and transformation are necessary steps before operational data can be used in a simulator.

A key challenge is the alignment between operational data and infrastructure data. A train record may indicate that a train travelled from one station to another, but the infrastructure model must contain a valid path between these locations. If the infrastructure is incomplete or if station identifiers do not match, the route cannot be generated. This challenge is particularly important in the micro-level model, where route generation depends on detailed track connectivity.

For this thesis, operational data is mainly used to generate train trips and simulation scenarios. In the macro-level model, real operational data can be more easily mapped to station-to-station routes. In the micro-level model, the same data requires a more detailed routing process because station-level movements must be translated into sequences of track segments and platform stops. This explains why data alignment and route feasibility are central issues in the construction of a railway digital twin.

2.3 Railway Simulators

Simulation plays an important role in the analysis, planning, and evaluation of railway systems. Since railway operations involve many interacting components, it is often difficult to test new strategies directly on the real network. Simulators provide a controlled environment in which infrastructure layouts, timetables, train movements, disruptions, and operational scenarios can be evaluated before being applied in practice.

In the context of DTRs, simulation is even more important. A digital twin requires a virtual environment able to reproduce the behaviour of the physical system with sufficient accuracy. This does not necessarily mean that every detail of the real railway network must be represented from the beginning. Rather, the simulation model must be consistent with the objective of the study. For example, a high-level model may be sufficient for analysing station-to-station movements, while a detailed infrastructure model is required to study switches, platforms, signalling constraints, and local conflicts.

Railway simulators can be classified according to several criteria. A first distinction concerns the level of granularity. Macroscopic simulators represent traffic at an aggregated level, often using stations, lines, flows, or timetable events. Microscopic simulators represent individual trains and infrastructure elements in more detail. A second distinction concerns the modelling paradigm. Some tools focus mainly on operational simulation, others on infrastructure design, timetable planning, capacity analysis, or real-time traffic management. Finally, simulators may differ in terms of openness, extensibility, data requirements, interoperability, and support for automated workflows.

2.3.1 Overview of Existing Simulators

Several simulation platforms exist for railway and transport systems. Some are specifically designed for railways, while others are general-purpose or multimodal traffic simulation environments that can be adapted to rail use cases.

Table 2.1: Comparative analysis of railway and transport simulation tools

Simulator	Type	Main focus	Main strengths	Main limitations	Relevance for this thesis
OpenTrack	Microscopic railway simulator	Railway operation, timetable evaluation, capacity and infrastructure analysis	Detailed railway-specific modelling; supports infrastructure, rolling stock, timetables and signalling constraints; provides train graphs, occupation diagrams and operational statistics	Requires detailed and well-structured railway data; less suited for fully custom open data pipelines; not primarily designed as an open experimental framework	Highly relevant as a reference railway simulator, but less suitable for the implemented data-driven and reproducible pipeline
RailSys	Professional railway planning and simulation suite	Capacity analysis, timetable planning, travel time computation, headway analysis and ETCS-related studies	Strong railway-specific capabilities; designed for infrastructure managers and railway operators; supports detailed operational and capacity studies	Commercial/professional environment; limited openness for academic reproducibility; requires detailed infrastructure and operational input data	Relevant for comparison with professional railway tools, but less aligned with the objective of building an open and programmable prototype
OSRD	Open-source railway-specific platform	Railway infrastructure design, capacity analysis, timetabling, simulation and short-term path requests	Open-source; railway-oriented by design; supports infrastructure and timetable workflows; suitable for capacity and pathfinding studies	More oriented toward railway design and planning workflows; integration with a custom heterogeneous data-processing pipeline may require additional modelling effort	Relevant because it is open-source and railway-specific, but SUMO was preferred for its flexibility, scriptability and simulation workflow control
SUMO	Open-source microscopic traffic simulator	Multimodal traffic simulation, route generation, demand modelling, public transport and railway simulation	Open-source and highly scriptable; supports XML-based network generation, routing tools, outputs, TraCI interaction and large-scale automated workflows; includes railway-specific features such as rail edges, train stops and rail signals	Not originally designed as a railway-only simulator; detailed signalling, interlocking, dispatching and platform assignment require additional modelling effort	Selected as the main simulation platform because it enables full control over network generation, trip generation, execution and output processing
AnyLogic / MATSim	General-purpose or transport simulation platforms	Agent-based simulation, multimodal mobility, large-scale transport demand modelling	Useful for multimodal and agent-based transport scenarios; suitable for studying passenger behaviour, demand patterns or large-scale mobility systems	Limited native support for detailed railway infrastructure constraints such as switches, interlocking, block occupation and platform-level operations	Useful as general transport simulation references, but less appropriate for the railway infrastructure reconstruction objectives of this thesis

The choice of a railway simulator depends on the intended use case. For a digital twin foundation, several criteria are important. First, the simulator must be able to represent the railway infrastructure at the required level of detail. Second, it must support train route generation and simulation. Third, it should be compatible with available data sources. Fourth, it should allow outputs to be extracted and compared with real operational data. Finally, for academic research, reproducibility and extensibility are important criteria.

Specialised railway simulators such as OpenTrack and RailSys provide strong railway-specific capabilities. They are designed to represent timetables, train movements, signalling systems, capacity constraints, and detailed infrastructure. This makes them highly relevant for professional railway planning and operational analysis. Their main advantage is that they directly encode railway concepts that are often simplified in general-purpose traffic simulators. However, they may also require detailed infrastructure and timetable data, and their integration into a fully custom data pipeline may be less straightforward depending on licensing, formats, and available interfaces.

OSRD offers an interesting alternative because it is open-source and railway-specific. It directly targets infrastructure design, capacity analysis, timetabling, and simulation. This makes it particularly relevant for railway planning workflows. However, its architecture and data model are more directly oriented toward railway design and pathfinding, whereas the objective of this thesis is to investigate how heterogeneous railway data can be transformed into simulation-ready artefacts and executed within a flexible traffic simulation environment.

SUMO occupies a different position. It is not as railway-specialised as OpenTrack, RailSys, or OSRD, but it provides a highly flexible and programmable simulation environment. Its main strengths are openness, extensibility, scalability, scriptability, and the availability of many tools for network generation, routing, demand generation, output processing, and online interaction. These characteristics make SUMO suitable for experimental research where the objective is not only to run a simulation, but also to generate the simulation network automatically from data, modify scenarios programmatically, and analyse outputs with custom scripts.

For this thesis, SUMO was selected because it allows the complete simulation workflow to be controlled. The network can be generated from XML files, routes can be created automatically, simulations can be executed through command-line tools, and outputs can be parsed and transformed into datasets suitable for analysis. This level of control is essential for comparing macro-level and micro-level railway representations and for evaluating the feasibility of integrating real operational data into a simulation pipeline.

Nevertheless, the use of SUMO also introduces limitations. Because SUMO was originally developed for general traffic simulation, some railway-specific aspects require additional modelling effort. Detailed railway signalling, interlocking, platform assignment, train dispatching, and timetable regulation are not automatically available unless they are explicitly represented in the input files or controlled through additional logic. Therefore, SUMO is well suited for building a flexible simulation-oriented digital twin foundation, but it does not remove the need for careful railway data modelling.

2.3.2 Limitations and Gaps in Current Simulators

Existing simulators provide important capabilities, but several limitations remain when the objective is to construct a railway digital twin from heterogeneous real-world data. A first limitation concerns data integration. Many simulators assume that the infrastructure and demand inputs are already available in a consistent and simulator-compatible format. In practice, railway data often comes from multiple sources, with different identifiers, geometries, levels of detail, and temporal resolutions. Transforming these datasets into valid simulation inputs remains a major challenge.

A second limitation concerns the balance between realism and scalability. Detailed railway simulators can represent signalling, switches, platforms, and operational conflicts, but this usually requires high-quality infrastructure data and careful configuration. Simpler models are easier to generate and scale, but they omit important operational constraints. This creates a trade-off between macroscopic representations, which are easier to use with large datasets, and microscopic representations, which are more realistic but more fragile.

A third limitation concerns reproducibility and openness. Some railway simulation tools are commercial or require proprietary workflows. This can be appropriate for industrial applications, but it may limit reproducibility in academic research. Open-source tools such as SUMO and OSRD reduce this barrier, but they still require significant preprocessing and modelling work before real railway data can be used.

A final limitation concerns the connection between simulation and operational data. Many simulation studies focus either on infrastructure modelling or on timetable evaluation, but fewer provide an end-to-end workflow that starts from raw infrastructure and punctuality data, generates a simulation-ready railway network, executes train movements, and compares simulation outputs with real observations. This gap directly motivates the work presented in this thesis.

2.3.3 SUMO: General Features

SUMO is an open-source traffic simulation suite designed to model large-scale transport systems. It is microscopic, meaning that each vehicle is represented individually, and continuous, meaning that vehicle movements evolve over simulation time rather than being represented only as discrete events. SUMO supports different transport modes, including cars, buses, pedestrians, bicycles, public transport, and rail vehicles [22, 7].

A SUMO simulation is built from several types of input files. The network file describes the topology of the simulated environment, including nodes, edges, lanes, connections, and permissions. Route files define vehicles, vehicle types, departure times, and paths. Additional files may define stops, detectors, traffic lights, calibrators, public transport stops, rerouting rules, or other simulation objects. A configuration file then specifies which inputs must be loaded and which outputs should be generated.

SUMO provides several tools for scenario creation and manipulation. The `netconvert` tool converts network descriptions into SUMO network files. It can import networks from multiple formats, including OpenStreetMap and manually defined XML files. The `duarouter` tool computes routes from trips or origin-destination information. The `randomTrips.py` script can generate synthetic travel demand. The `netedit` tool provides a graphical interface for editing networks. These tools make SUMO particularly suitable for building simulation scenarios through automated pipelines.

SUMO also provides a rich set of outputs. Depending on the configuration, it can generate information about completed trips, stops, vehicle trajectories, edge-based statistics, lane-based statistics, emissions, delays, and simulation summaries. These outputs are useful because they can be transformed into structured datasets and compared with real observations. In this thesis, outputs such as trip information, stop information, and summary statistics are particularly important because they allow simulated train movements to be evaluated against operational data.

Another important feature is `TraCI`, the Traffic Control Interface. TraCI allows external programs, typically written in Python, to interact with a running SUMO simulation. Through TraCI, it is possible to retrieve the state of the simulation, control vehicles, modify traffic lights, reroute vehicles, or implement custom control logic. This is especially relevant for digital twin applications because it provides a mechanism through which simulation can become interactive rather than purely offline.

2.3.4 SUMO Backend Architecture

SUMO can be understood as a modular simulation ecosystem. Its architecture is organized around a set of command-line applications, input/output formats, libraries, and interfaces that together support the complete simulation workflow.

At the core of the system is the simulation engine, executed through `sumo` or `sumo-gui`. The command-line version is used for automated or large-scale experiments, while the graphical version provides visualization and debugging capabilities. Both versions rely on the same simulation principles and load the same core input files. This separation between simulation execution and graphical visualization is useful because it allows scenarios to be developed visually and then executed automatically in scripts.

Network generation is handled by dedicated tools such as `netconvert` and `netgenerate`. In the context of this thesis, `netconvert` is especially important because it transforms node and edge definitions into a complete SUMO network. This step includes the construction of lanes, connections, permissions, and network topology. For railway simulation, this conversion step is critical because routing feasibility depends on the correctness of the generated topology.

Demand generation and routing are handled by tools such as `randomTrips.py` and `duarouter`. `randomTrips.py` can generate synthetic trips according to user-defined parameters, while duarouter computes valid paths over the network. In railway scenarios, these tools can be used to generate artificial train movements or to convert reconstructed operational trips into executable routes.

SUMO also includes Python libraries such as `sumolib`, which allows SUMO networks and XML files to be inspected and manipulated programmatically. This is important for research workflows because it makes it possible to build custom preprocessing, validation, and post-processing scripts. Together with `TraCI`, these libraries provide a bridge between SUMO and external data-processing environments.

The backend architecture is therefore particularly suitable for the methodology of this thesis. The proposed pipeline relies on Python and Apache Spark for data processing, generates XML files compatible with SUMO, uses SUMO tools for network conversion and routing, and then parses simulation outputs for analysis. This modular workflow allows each step to be inspected, replaced, or improved independently.

2.3.5 SUMO For Railway Simulation

Although SUMO was initially developed for urban mobility simulation, it includes specific functionalities for railway simulation. Railway networks can be imported from supported formats or built manually by defining edges with railway permissions. When importing railway data from `OpenStreetMap`, railway edges can be configured with appropriate vehicle classes such as rail. When building networks from scratch, railway behaviour can be obtained by assigning the *allow* = "*rail*" attribute to the corresponding edges [6].

Railway simulation in SUMO differs from road simulation in several important ways. First, railway tracks should usually be represented as distinct edge elements, especially when parallel tracks exist. Unlike road lanes, which may represent parallel lanes of the same road segment, parallel railway tracks often have separate operational roles and routing possibilities. SUMO documentation therefore recommends representing parallel tracks as distinct edges rather than as multiple lanes of a single edge [6].

Second, railway directionality must be handled explicitly. In real railway systems, tracks may physically support traffic in both directions, but operational practice often assigns preferred directions. In SUMO, bidirectional railway usage is represented by two opposite edges with reversed geometries. This makes directionality explicit in the network topology and allows routing preferences or restrictions to be defined. This point is important for micro-level railway modelling because missing or incorrect directionality may prevent valid train routes from being found.

Third, trains can be associated with stops. In SUMO, railway stops can be represented using `trainStop` elements. A train route may include scheduled stops with attributes such as duration or until time. This is essential for representing passenger railway operations because trains do not simply travel from origin to destination, they stop at intermediate stations according to a timetable.

Fourth, SUMO provides rail signal functionalities. Rail signals can regulate access to railway blocks, support schedule constraints, and prevent certain deadlock situations. The documentation also describes moving-block behaviour and schedule constraints that can enforce train ordering at signals. These mechanisms show that SUMO includes railway-specific control features, although their effective use requires the input network, signals, routes, and schedules to be modelled correctly [6].

Lastly, SUMO provides railway-specific tools and outputs. For example, rail simulation can generate outputs related to signalling blocks and driveway occupation. Tools such as `checkStopOrder.py`, `scheduleStats.py`, and `checkReversals.py` can be used to analyse railway schedules, compare simulated train behaviour with schedule expectations, and detect problematic train routes. These tools are relevant for validating railway scenarios and identifying issues in route generation.

Despite these features, SUMO railway simulation has limitations. It does not automatically reconstruct railway infrastructure from raw operational data. It also does not automatically infer complete platform assignments, switch logic, or signalling constraints from incomplete datasets. Therefore, the quality of a SUMO railway simulation depends strongly on the preprocessing pipeline used to generate the network, stops, routes, and additional files. This is precisely where the contribution of this thesis is located. The work investigates how Belgian railway infrastructure and punctuality data can be transformed into SUMO-compatible macro-level and micro-level railway simulations.

2.4 Simulation Calibration

Simulation calibration is the process of adjusting model parameters so that simulation outputs reproduce observed real-world behaviour as closely as possible. In transport simulation, calibration is necessary because a simulator is never a perfect copy of reality. It relies on assumptions, parameters, simplifications, and incomplete data. Without calibration, the simulation may run correctly from a technical perspective but still produce unrealistic travel times, flows, delays, or capacity usage.

In microscopic traffic simulation, calibration often concerns parameters related to vehicle behaviour, such as acceleration, deceleration, reaction time, car-following parameters, lane-changing behaviour, route choice, and departure distributions. In railway simulation, calibration may concern different elements: running times, dwell times, train acceleration and braking behaviour, platform stopping times, route choices, dispatching rules, delay distributions, and capacity constraints.

Several calibration approaches exist. Manual calibration consists of adjusting parameters based on expert knowledge and comparing simulation outputs with reference data. This approach is simple but can be subjective and difficult to scale. Optimization-based calibration defines an objective function that measures the difference between simulated and observed data, then searches for parameter values that minimize this difference. Metaheuristic methods, such as genetic algorithms, simulated annealing, or tabu search, can be used when the parameter space is complex. Data-driven calibration methods use statistical or machine learning techniques to estimate model parameters from observed data.

A calibration process usually requires three elements. First, reference data must be available. In railway applications, this may include planned and observed arrival times, departure times, dwell times, delays, platform usage, or track occupation data. Second, comparable simulation outputs must be produced. Third, evaluation metrics must be defined. These metrics may include mean absolute error, root mean square error, delay distribution similarity, travel time error, punctuality indicators, or the difference between observed and simulated traffic volumes.

SUMO includes calibration-related mechanisms, especially for road traffic. For example, calibrators can insert or remove vehicles to match observed flow and speed values over specific time intervals [5]. More generally, SUMO can be integrated into external calibration workflows because simulations can be launched automatically, parameters can be modified through scripts, and outputs can be analysed programmatically. This makes it possible to build iterative calibration loops around the simulator.

In this thesis, calibration is not the main focus. The objective is not to tune the simulator until it reproduces railway traffic perfectly, but rather to construct a simulation-ready railway model and evaluate whether real railway data can be transformed into feasible SUMO scenarios. Nevertheless, calibration remains relevant for future work. Once the infrastructure and routing problems are solved, calibration could be used to adjust dwell times, running times, departure distributions, and delay behaviour so that the simulated railway system better matches historical punctuality data.

2.5 Research Gaps

The literature and existing tools show that railway simulation and digital twin technologies are both mature enough to support advanced applications, but several gaps remain when the objective is to build a data-driven railway digital twin simulation from heterogeneous real-world data.

The first gap concerns the transformation of heterogeneous railway data into simulation-ready models. Existing simulators often assume that infrastructure, timetable, and operational data are already available in compatible formats. However, real railway datasets are rarely directly usable. Infrastructure data may describe tracks, stations, platforms, and switches, while operational data may describe train movements only at the station level. These datasets must be cleaned, aligned, transformed, and validated before they can be used in simulation. This preprocessing step is often underestimated, even though it determines whether the simulation can run at all.

The second gap concerns the comparison between different levels of railway representation. A macro-level representation is easier to generate and can support large-scale station-to-station simulations, but it ignores detailed infrastructure constraints. A micro-level representation is more realistic because it explicitly represents tracks, switches, platforms, and routing constraints, but it is also more sensitive to data quality and topological inconsistencies. Existing works often focus on one level of abstraction, whereas this thesis explicitly studies both and analyses the trade-off between simplicity, scalability, realism, and feasibility.

The third gap concerns the integration of real operational data into railway simulation. Synthetic trips are useful for testing infrastructure connectivity, but they do not reproduce the actual structure of railway operations. Real punctuality data provides a more realistic basis for simulation, but it does not directly contain the complete physical route followed by each train. As a result, a route reconstruction process is required. This problem becomes particularly difficult at the micro level, where each station-to-station movement must be mapped onto a valid sequence of track segments and platform stops.

This thesis addresses these gaps by proposing and evaluating a data-driven pipeline for constructing SUMO-based railway simulations at two levels of abstraction. The macro-level model represents stations and station-to-station connections, allowing real train trajectories to be reconstructed more easily from operational data. The micro-level model represents tracks, switches, platforms, and routing structures, making it possible to study the challenges of detailed railway simulation. By comparing these two approaches, the work contributes to understanding the practical requirements, limitations, and trade-offs involved in building simulation-oriented railway digital twins.

Chapter 3

Proposed Methodology

This chapter presents the methodology proposed in this thesis to construct a simulation-oriented digital twin of a railway network. The objective is to define and implement the foundational components required to transform heterogeneous railway datasets into simulation-ready models. These components include infrastructure reconstruction, operational data integration, trip generation, simulation execution, and result validation.

The proposed methodology is structured around two complementary levels of representation. The first one is a macro-level model, in which the railway network is represented as a station-to-station graph. This model provides a simplified but scalable representation of the railway system and is mainly used to reconstruct and simulate train movements from real operational data. The second one is a micro-level model, in which the railway infrastructure is represented in greater detail through tracks, switches, platforms, and operational segments. This model aims to move closer to the physical structure of the railway network and to evaluate whether more detailed train simulations can be generated from the available data.

This dual-level approach makes it possible to study the trade-off between simplicity and realism. The macro-level model is easier to construct, more robust for route reconstruction, and better suited for validating the end-to-end workflow. However, it cannot represent local infrastructure constraints inside stations. The micro-level model provides a richer infrastructure description, but it also introduces additional challenges related to data completeness, platform assignment, switch detection, route feasibility, and simulation complexity. By developing both approaches, the methodology evaluates how different levels of abstraction influence the construction of a railway digital twin.

3.1 Positioning of This Work

The state of the art shows that railway digital twins and railway simulation tools provide a relevant foundation for analysing railway systems. They make it possible to combine infrastructure data, operational data, simulation models, and analytical services within a common framework. However, the literature also highlights that building an executable railway simulation from real-world data remains difficult, mainly because railway data is heterogeneous, incomplete, and represented at different levels of detail.

A first challenge concerns the transformation of raw railway datasets into simulation-ready inputs. Existing simulators often assume that infrastructure, timetable, and operational data are already available in consistent and compatible formats. In practice, infrastructure data may describe stations, tracks, switches, platforms, or geometries, while operational data may only describe train passages at station level. Aligning these datasets is therefore a central step in the construction of a railway digital twin.

A second challenge concerns the level of representation. A macro-level model is easier to generate and more robust for large-scale station-to-station simulation, but it ignores detailed infrastructure constraints. A micro-level model is closer to the physical infrastructure because it represents tracks, platforms, switches, and routing possibilities, but it is also more sensitive to missing data, incorrect topology, and routing inconsistencies.

This thesis is positioned at the intersection of these issues. Its objective is not to develop a complete real-time railway digital twin or to replace professional railway simulators, but to study the foundational steps required to build simulation-ready railway models from heterogeneous data. The proposed methodology therefore develops two complementary representations: a macro-level model, used to validate the integration of real operational data into simulation, and a micro-level model, used to investigate the challenges of detailed infrastructure reconstruction.

By comparing these two approaches, this work addresses the gap identified in the literature concerning end-to-end workflows for railway digital twin construction. The contribution lies in the design, implementation, and evaluation of a data-driven pipeline able to transform infrastructure and punctuality data into SUMO-compatible railway simulation scenarios.

3.2 Framework

The proposed framework follows a data-driven pipeline that transforms raw railway datasets into SUMO-compatible simulation artefacts. The general architecture can be divided into five main stages: data collection, data preprocessing, infrastructure modeling, trip generation, and simulation-based evaluation. These stages are common to both the macro-level and micro-level approaches, but their implementation differs according to the level of detail required by each model.

The first stage consists of collecting the available railway data. Two main categories of data are considered: infrastructure data and operational data. Infrastructure data describes the physical and topological structure of the railway network, including stations, station-to-station connections, tracks, platforms, and switches. Operational data describes train movements, timetables, arrival and departure times, and punctuality information. These datasets provide the basis for constructing the virtual representation of the railway system.

The second stage corresponds to data preprocessing. Since railway data is heterogeneous, it must be cleaned, standardized, and transformed before it can be used for simulation. This includes harmonizing station identifiers, processing geographical coordinates, filtering the data according to the selected simulation scope, and converting temporal information into simulation-compatible time values. This step is essential because inconsistencies between infrastructure and operational datasets can prevent route reconstruction or lead to invalid simulations.

The third stage is infrastructure modeling. In the macro-level model, the infrastructure is represented as a graph in which stations correspond to nodes and direct railway connections correspond to edges. This representation is suitable for station-to-station simulation and for reconstructing train trajectories from punctuality data. In the micro-level model, the infrastructure is reconstructed at a finer level of detail. Individual tracks are represented explicitly, switches are detected from network topology, platforms are matched to nearby track segments, and the network is segmented when necessary to improve the operational granularity of the model.

The fourth stage is trip generation. In the macro-level model, train trips are reconstructed from real punctuality records. Each train journey is obtained by grouping station records by train identifier and operating day, then sorting them according to their planned times. The resulting station sequences are converted into SUMO routes. In the micro-level model, two trip generation strategies are considered. The first one generates random trips constrained by the detected platforms and network topology. The second one attempts to reconstruct real train journeys from operational data and map them onto the detailed infrastructure.

The final stage is simulation and evaluation. Once the network and routes have been generated, the scenario is executed in SUMO. The simulation produces output files describing train movements, stops, travel times, and summary statistics. These outputs are then processed and compared with the available reference data. The evaluation focuses on dataset coverage, infrastructure modeling quality, and the feasibility of the generated simulations. The framework therefore provides both a technical implementation pipeline and an evaluation methodology for assessing the quality of the generated railway digital twin.

3.3 Why SUMO?

The choice of SUMO is motivated by the objective of this thesis: generating railway simulation scenarios automatically from heterogeneous datasets. Several railway simulation tools exist, including OpenTrack, RailSys, and OSRD. These tools provide strong railway-specific functionalities and are highly relevant for professional planning, timetable evaluation, and capacity analysis. However, they often require detailed and well-structured railway input data, whereas this work focuses on building a flexible and reproducible pipeline from raw infrastructure and operational datasets.

SUMO was selected because it is open-source, scriptable, and compatible with automated data-processing workflows. Its XML-based input structure makes it possible to generate networks, routes, stops, and configuration files directly from Python scripts. It also provides command-line tools which are useful for converting networks, generating trips, and computing executable routes.

Another advantage of SUMO is that it can support both levels of representation considered in this thesis. At the macro level, stations can be represented as nodes and railway connections as edges. At the micro level, railway infrastructure can be represented with rail edges, train stops, lanes, and more detailed route definitions. This flexibility makes SUMO suitable for comparing simplified station-to-station simulation with more detailed infrastructure-based simulation.

SUMO also produces structured outputs, such as trip information, stop information, and simulation summaries, which can be processed after execution and compared with operational data. In addition, tools such as sumolib and TraCI provide possibilities for future extensions involving more interactive or real-time digital twin functionalities.

Nevertheless, SUMO is not a railway-only simulator. Detailed signalling, interlocking, dispatching, platform assignment, and conflict management are not automatically inferred from raw data. These elements must be explicitly modelled or added through additional logic. For this thesis, this limitation is acceptable because the goal is precisely to study how railway data can be transformed into simulation-ready models. SUMO therefore provides a flexible platform for developing and evaluating the proposed railway digital twin prototype.

3.4 Data Available

The proposed methodology relies on two main types of datasets: infrastructure data and operational data. Infrastructure data is used to construct the virtual representation of the railway network, while operational data is used to generate train movements and evaluate whether real railway activity can be reproduced in simulation. The use of both categories is essential because a railway digital twin must represent not only the physical structure of the network, but also the way trains actually operate on it.

The data used in this thesis comes from Infrabel Open Data Portal And DTSC who has access to private data directly from Infrabel. These datasets are not directly simulation-ready. They must be cleaned, aligned, transformed, and sometimes approximated before they can be converted into SUMO artefacts. The available data also differs depending on the level of representation. The macro-level model mainly uses stations and station-to-station connections, while the micro-level model requires more detailed information about tracks, platforms, and switches.

3.4.1 Infrastructure Data

Infrastructure data corresponds to the static elements of the railway network. It includes information about stations, connections between stations, railway track geometries, platform locations, and switch-like topological structures. In this thesis, infrastructure data is used to generate the SUMO network on which train movements are simulated.

The use of infrastructure data differs between the macro-level and micro-level approaches. The macro-level model only requires stations and direct connections between them. It therefore focuses on the general connectivity of the railway network. The micro-level model requires more detailed infrastructure elements and aims to represent the physical topology of the railway network more closely.

Macro-Level Data

The macro-level infrastructure data represents the railway system as a graph. In this graph, stations are modeled as nodes and direct railway connections between stations are modeled as edges. This abstraction does not include detailed elements such as platforms, switches, signals, or internal station layouts. Instead, it focuses on preserving the overall topology of the railway network.

This level of data is particularly useful for reconstructing train movements from station sequences. Since the operational data contains train passages through stations, the macro-level representation makes it possible to map these station sequences directly onto a simulation network. The objective is therefore not to reproduce detailed infrastructure conflicts, but to build a robust and scalable representation of railway circulation between stations.

Dataset	Example attributes	Reference value	Role
Stations	Station identifier, station name, latitude, longitude	559 stations	Used to create SUMO nodes representing railway stations
Stations Connections	Origin station, destination station, distance, geometry	1385 tracks	Used to create SUMO edges between neighbouring stations
Selected Simulated Area	List of selected stations or station chain	User-defined	Used to restrict the simulation to a coherent part of the network

Table 3.1: Macro-Level Infrastructure Data Overview

The macro-level data therefore provides the minimal infrastructure required to simulate trains from one station to another. Its main advantage is that it reduces the complexity of the railway system while keeping the information needed to reconstruct real train trajectories.

Micro-Level Data

The micro-level infrastructure data provides a more detailed representation of the railway network. It considers physical railway elements such as track segments, platforms, and switches. This level of representation is required when the objective is to simulate train circulation closer to the real infrastructure.

Track data is used to generate the base railway graph. Each track segment contains geometric and topological information that makes it possible to construct the SUMO network. Platform data is used to identify where trains can stop inside stations. Since platforms must be associated with SUMO lanes, a matching step is required to link each platform to a nearby track segment. Switches are not used only as isolated objects, but also as topological indicators that help describe how track segments are connected.

An example of the type of information used at the micro level is shown in Table 3.2.

Dataset	Example attributes	Reference value	Role
Tracks segments	Track identifier, geometry, start point, end point, length, geometry	25068 tracks	Used to construct detailed SUMO railway edges
Platforms	Station name, platform number, associated area	1560 platforms	Used to create train stops in the SUMO additional file
Switches	Node identifier, coordinates, connected track segments	10379 switches	Used to represent branching points and support routing through the network

Table 3.2: Micro-Level Infrastructure Data Overview

3.4.2 Operational Data

Operational data describes train movements across the railway network. In this thesis, this data is mainly used to reconstruct train journeys and to generate simulation routes. Each record corresponds to the passage of a train at a given station and contains temporal information related to the planned and observed operation of the service.

The operational dataset is essential because it provides a connection between the static infrastructure and real railway activity. Without operational data, the generated network would only represent the physical railway system, but it would not contain realistic train movements. By using train identifiers, station identifiers, and planned times, it becomes possible to reconstruct ordered sequences of stops for each train.

An example of the type of information contained in the operational data is shown in Table 3.3.

Dataset	Main attribute groups	Example attributes	Reference value	Role
Punctuality dataset	Train identification, station passage, timetable information, observed operation, delay information	Train identifier, operating date, station identifier, planned arrival time, planned departure time, observed arrival/departure time, arrival delay, departure delay	Depends on selected time window	Used to reconstruct real train journeys, generate simulation routes, and compare simulated operations with observed railway movements

Table 3.3: Operational Data Overview

At the macro level, the operational data can be mapped relatively directly onto the station-to-station graph. For each train and each operating day, the records are grouped and sorted chronologically to obtain a sequence of visited stations. This sequence is then transformed into a SUMO route.

At the micro level, the same operational data is more difficult to use because each station stop must be associated with a specific platform or track segment. In addition, the route between two consecutive stations must be computed through the detailed infrastructure graph. The operational data therefore remains the same in principle, but its integration becomes more constrained when the level of infrastructure detail increases.

Overall, the available data provides the foundation for constructing the proposed railway digital twin. Infrastructure data defines the network to be simulated, while operational data defines the train movements to be reproduced. The distinction between macro-level and micro-level data is central to the methodology, because it determines the type of simulation that can be generated and the complexity of the preprocessing steps required before execution in SUMO.

Chapter 4

Implementation

4.1 Macro-Level Modeling

The macro-level model constitutes the first operational implementation of the railway digital twin developed in this work. Its primary objective is to generate a simulation-ready representation of the Belgian railway network while maintaining a level of abstraction that remains computationally manageable. Unlike the micro-level model presented later in this thesis, which aims to reproduce the detailed infrastructure of railway stations and track layouts, the macro-level model focuses on the connectivity between stations and the movement of trains across the network.

At this level of abstraction, railway stations are represented as nodes and railway connections between stations are represented as edges. Detailed infrastructure components such as switches, signaling systems, individual tracks, and platform assignments are intentionally omitted. This simplification significantly reduces the complexity of the generated network while preserving its overall topology and operational structure.

The implementation of the macro-level model serves several purposes. First, it validates the complete workflow required to construct a railway digital twin, from raw infrastructure and operational data to simulation outputs. Second, it provides an initial environment for evaluating trip generation methods based on real railway operations. Finally, it establishes a baseline model against which the more detailed micro-level implementation can later be compared.

The implementation was developed in Python and relies on Apache Spark for large-scale data processing and SUMO for railway simulation. The complete workflow consists of four main stages: data processing, railway network generation, trip generation, and simulation execution.

4.1.1 Data Processing Pipeline

The generation of the macro-level railway model begins with the construction of a data processing pipeline responsible for transforming heterogeneous railway datasets into a format compatible with SUMO. Railway data originates from a unique source but exhibits significant differences in structure, granularity, and purpose. Consequently, a preprocessing phase is required before any simulation artefacts can be generated.

Three categories of data are used within the macro-level model:

- **Railway station data:** contains information describing the geographical location and identification of Belgian railway stations.

- **Railway connectivity data:** provides information regarding the direct railway links connecting stations.
- **Railway punctuality data:** operational records describing the observed movements of trains across the railway network.

The preprocessing workflow follows a classical Extract-Transform-Load (ETL) approach. During the extraction phase, raw datasets are loaded into Apache Spark dataframes. Spark was selected because of its ability to efficiently process large datasets while providing a unified interface for data transformation operations. Although the macro-level simulation itself does not require distributed execution, the volume of punctuality records makes Spark particularly suitable for preprocessing tasks.

Once loaded, the datasets undergo a transformation phase during which inconsistencies are corrected and the data structure is standardized. Particular attention is paid to geographical information. Infrastructure datasets and operational datasets do not always use identical coordinate representations, resulting in small discrepancies between station locations and railway line geometries. Coordinate normalization procedures are therefore applied to ensure consistency between datasets.

A filtering phase is subsequently performed to restrict the simulation to the geographical area and time period selected by the user. The simulation generator allows a subset of stations to be specified as input. In addition to these stations, neighboring stations required to preserve network connectivity are automatically included in the simulation. This approach reduces computational complexity while ensuring that realistic routes can still be generated.

The output of the preprocessing pipeline consists of a collection of cleaned datasets containing standardized station information, station-to-station railway connections, and train circulation records. These datasets constitute the foundation upon which the simulation network is generated.

4.1.2 Railway Network Generation

Once the preprocessing phase has been completed, the railway network can be generated. The objective of this stage is to transform the cleaned infrastructure datasets into a graph representation that can subsequently be converted into a SUMO railway network.

The process begins with the loading of station information. Each station is uniquely identified by its official identifier and associated with a geographical position. Internally, station information is stored in dictionary-based structures that provide efficient access during subsequent processing stages. For each station, the implementation stores its identifier, name, and geographical coordinates.

After station loading, railway links between stations are reconstructed from the station-to-station dataset. Each railway connection contains the identifier of the departure station, the identifier of the arrival station, the physical distance separating the two stations, and the geographical shape describing the railway corridor.

Unlike a purely abstract graph representation, the implementation preserves the geometry of railway lines whenever this information is available. Railway corridors are therefore represented as polylines rather than simple straight segments. This decision improves the visual realism of the generated SUMO network while preserving the simplicity of the macro-level model.

The resulting railway network can be interpreted as a directed graph $G = (V, E)$, where the set of vertices V corresponds to railway stations and the set of edges E corresponds to direct railway connections between stations.

Once the graph has been constructed, a series of XML files compatible with SUMO are generated automatically. These files include the node definitions, edge definitions, stop definitions, train definitions, and simulation configuration files required to execute the simulation.

The generated node file contains all stations included in the simulation. Each station is represented as a SUMO node associated with a unique identifier and geographical coordinates. Similarly, the edge file contains the railway links connecting stations and specifies the geometry and characteristics of each connection.

The final step consists of converting these files into a SUMO network using the netconvert utility. This tool automatically transforms the graph representation into a simulation-ready railway network. The resulting network preserves the geographical distribution of stations and railway corridors while remaining significantly simpler than the actual Belgian railway infrastructure.

This abstraction offers several advantages. The resulting network is easy to generate, computationally efficient, and sufficiently realistic for validating trip generation and operational workflows. However, it also introduces limitations. Since stations are represented by a single node, infrastructure-level phenomena such as route conflicts, switch occupation, signal interactions, and platform assignment cannot be reproduced accurately.

4.1.3 Trip Generation

Once the railway network has been generated, train movements must be created in order to execute the simulation. Rather than relying exclusively on synthetic traffic generation, the proposed methodology aims to reconstruct train movements from real operational data.

The punctuality dataset contains records describing successive train passages through railway stations. Each record includes a train identifier, a station identifier, and temporal information such as planned departure times and observed departure times. Although these records provide valuable operational information, they do not directly describe complete train journeys.

A reconstruction process is therefore required to transform isolated station observations into coherent train routes. For each train and operating day, records are first sorted chronologically according to their planned departure time. The next station visited by the train is then identified using window-based operations implemented in Apache Spark.

This procedure allows consecutive station pairs to be extracted and transformed into station-to-station movements. The resulting dataset can be interpreted as a collection of train trajectories describing the progression of trains through the railway network.

For each reconstructed movement, a simulation departure time must be computed. Since SUMO internally represents time as the number of seconds elapsed since the beginning of the simulation, departure times observed in the punctuality dataset are converted into simulation timestamps. This conversion is performed by calculating the difference between the planned departure time and the simulation start date selected by the user.

The reconstructed train movements are subsequently aggregated into complete train journeys. Each journey contains the train identifier, the sequence of visited stations, and the associated departure times. These journeys are then translated into SUMO route definitions.

The use of real operational data offers several advantages over purely random traffic generation. First, it preserves realistic traffic density patterns. Second, it reproduces actual station usage frequencies and train departure distributions. Finally, it provides a more representative basis for evaluating the behaviour of the digital twin, since the generated traffic reflects real railway operations rather than artificial assumptions.

4.1.4 Simulation

After the generation of the macro-level railway network and the reconstruction of train routes from operational data, the simulation scenario can be executed within SUMO. The objective of this step is to validate the ability of the generated model to reproduce railway movements at a station-to-station level while keeping the simulation computationally manageable.

The macro-level simulation is not intended to reproduce the full Belgian railway network in a single scenario. Although the generated network can theoretically include a large number of stations, simulating the complete national traffic would introduce a high level of complexity and would make the interpretation of the results more difficult. For this reason, the simulation is restricted to selected parts of the network. These parts correspond to chains of neighbouring stations where the train traffic remains moderate. In practice, this means that the user selects a set of stations to be included in the simulation, and the generation pipeline constructs the corresponding local railway subnetwork.

This selection strategy is particularly important for the macro-level model. Since stations are represented as single nodes and complex station layouts are not explicitly modelled, highly congested stations are avoided. Major stations with dense traffic, multiple platforms, many intersecting services, and complex internal routing would require infrastructure details that are not represented at this level of abstraction. By focusing instead on station chains with lower traffic intensity, the simulation remains coherent with the assumptions of the macro-level model. The simplified representation is therefore used in a context where local infrastructure conflicts are less dominant and where station-to-station train movements can still be meaningfully simulated.

A second advantage of this restriction is that real operational trajectories can be directly integrated into the simulation. Since the selected stations form a coherent and connected railway chain, the reconstructed train movements extracted from the punctuality data can be mapped onto the generated SUMO network. Each train route is defined by a sequence of stations and by departure times converted into simulation time. The resulting route file therefore reflects real observed railway operations rather than artificial movements.

The duration of the simulation is configurable. The user can specify the number of operating days to be included in the scenario. This parameter determines the time window used when filtering the operational data and therefore controls the number of train movements inserted into the simulation. A short simulation period can be used for debugging and validation, while a longer period can be selected to analyse traffic behaviour over multiple days.

Once the network, train routes, stop definitions, and simulation configuration files have been generated, the simulation can be launched directly in SUMO. During execution, trains circulate along the station-to-station graph according to the reconstructed timetable. Their movements are mainly governed by the generated routes, departure times, and travel times associated with the railway connections. The simulation produces standard SUMO output files, including trip information, stop information, and summary statistics. These outputs are then processed in order to obtain structured datasets that can be compared with the original operational data.

Several outputs are produced during simulation execution. The `tripinfo` output contains information regarding completed train journeys, including travel times and delays. The `stopinfo` output records stopping events occurring at stations. Finally, `summary` files provide aggregated information concerning the evolution of the simulation.

The post-processing stage reconstructs both scheduled and simulated arrival and departure times for each train and station. This conversion makes it possible to directly compare simulation results with real railway observations and therefore evaluate the validity of the generated digital twin.

In this way, the macro-level simulation acts as a first validation layer for the proposed railway digital twin. It does not aim to capture all local operational constraints of the railway infrastructure, but it demonstrates that a realistic simulation workflow can be constructed from real station connectivity and punctuality data. It also provides a scalable basis for analysing train movements on selected parts of the railway network before moving toward the more detailed micro-level model.

The complete workflow implemented in the macro-level model can therefore be summarized as follows:

1. Infrastructure and operational data are first preprocessed and standardized.
2. A station-to-station railway network is then generated and converted into a SUMO network.
3. Historical punctuality records are subsequently transformed into train routes.
4. These routes are simulated within SUMO.
5. The resulting outputs are finally converted into datasets suitable for analysis and validation.

4.1.5 Limitations

Although the macro-level model provides a functional and computationally efficient representation of the railway network, it remains limited by its high level of abstraction. Stations are represented as single nodes and railway connections as direct edges, which means that detailed infrastructure elements such as platforms, switches, signals, parallel tracks, and station-level routing constraints are not explicitly modelled. As a result, the model cannot accurately reproduce local operational phenomena such as platform occupation, route conflicts inside stations, or interactions between trains at complex junctions.

Another limitation concerns the scope of the simulation. The macro-level approach is mainly suitable for selected chains of neighbouring stations with moderate traffic intensity. It is therefore less appropriate for dense areas or major stations where the internal infrastructure plays a central role in traffic behaviour. Nevertheless, this simplified model remains useful as a first validation step for the overall digital twin workflow, especially for testing the integration of real operational data into a SUMO-based railway simulation.

4.2 Micro-Level Modeling

While the macro-level model provides a simplified representation of the railway network suitable for large-scale traffic simulations, it does not accurately represent the physical infrastructure required for detailed operational analyses. In particular, elements such as switches, platforms, parallel tracks, and complex station layouts are either simplified or omitted. To overcome these limitations, a micro-level railway network model was developed.

The objective of the micro-level modeling process is to reconstruct the railway infrastructure with a much higher level of spatial and topological accuracy by exploiting detailed infrastructure data together with operational information. Unlike the macro-level model, where stations are represented as single nodes connected through aggregated railway links, the micro-level model explicitly represents every individual track, switch, platform, and operational segment. This detailed representation enables realistic train routing, precise conflict detection, infrastructure capacity analysis, and the simulation of operational scenarios at station level.

The construction of the micro-level network follows a multi-stage processing pipeline. First, raw infrastructure data are collected from the available railway datasets and transformed into a consistent geographic representation. The resulting railway graph is then enriched through several dedicated algorithms responsible for identifying key operational elements of the infrastructure. These include automatic switch detection, platform identification, railway track segmentation, and the construction of a dual graph representation suitable for routing algorithms.

Once the infrastructure has been reconstructed, train trips are generated using either synthetic traffic scenarios or real operational data. These trips are subsequently integrated into the SUMO railway simulation environment, allowing trains to circulate on the generated infrastructure while respecting the physical constraints imposed by the detailed network topology.

Compared to the macro-level model, this approach significantly increases the realism of the generated digital twin. The explicit representation of railway assets enables more accurate train movements, better infrastructure utilization, and the possibility of evaluating local operational phenomena such as conflicts at junctions, platform occupancy, and the propagation of delays within complex stations.

The following sections describe each stage of the micro-level modeling process in detail, starting with the data processing pipeline before presenting the infrastructure extraction algorithms, trip generation methodology, simulation setup, and the main limitations of the proposed approach.

4.2.1 Data Processing Pipeline

The data processing pipeline follows a multi-stage Extract–Transform–Load (ETL) process. Infrastructure data is first collected from the available railway datasets and transformed into a unified format suitable for network reconstruction. The raw geographical information contains track geometries represented as sequences of coordinates describing the railway centerlines.

After data cleaning and normalization, several infrastructure extraction algorithms are applied. The first stage consists of identifying railway switches from the track topology. These switches constitute the connection points between different railway tracks and form the basis of the network graph.

The second stage focuses on platform detection. Stations are associated with nearby track segments using a proximity-based matching procedure that assigns platforms to the closest available tracks.

Once platforms have been matched, the network is analysed to identify situations where multiple stations are assigned to the same track segment. Such conflicts are resolved through a dedicated track segmentation process that increases the granularity of the infrastructure representation while preserving the original geometry.

The resulting infrastructure model is then converted into a graph representation that can be used for route generation and simulation. Finally, train trips are generated using either synthetic traffic scenarios or operational railway data before being integrated into the SUMO simulation environment.

4.2.2 Infrastructure Modeling

Switch Detection

The railway infrastructure is first represented as a graph in which track extremities correspond to nodes and track sections correspond to edges. For each node, the number of neighbouring track segments is computed.

Nodes connected to exactly two neighbouring segments are considered part of a continuous railway line and are therefore not classified as switches. In contrast, nodes connected to three or more neighbouring segments represent branching or merging points and are identified as switches. These locations correspond to operational infrastructure elements that allow trains to change routes within the network.

The resulting switch and edge datasets forms the foundation of the micro-level network model and is subsequently used for graph construction, route generation, and simulation.

Platform Detection

The infrastructure dataset provides the geographical position of railway stations as well as the number of platforms available at each station and their heights. However, this information is not directly linked to the track segments extracted from the railway network. Consequently, a matching procedure was developed to associate each station platform with the most relevant track segments.

The proposed approach relies exclusively on spatial proximity. For each station, the Euclidean distance between the station location and nearby track segments is computed. The candidate segments are then ranked according to their distance to the station centroid.

Let (N) denote the number of platforms associated with a given station. The algorithm selects the (N) closest track segments and assigns them to the station.

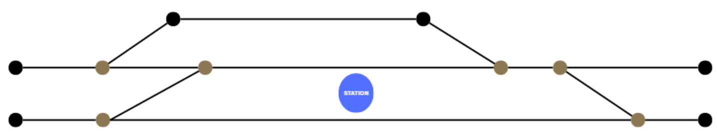

Figure 4.1: Example of platform detection initial setup

For each station, a research zone is defined around its coordinates. This research zone is a circular area with an adaptative radius (e.g., 100 meters) that encompasses the area where they are likely located. The radius of the research zone can be adjusted based on the density of platforms in the area, ensuring that it is large enough to capture all relevant track segments while minimizing the inclusion of irrelevant segments.

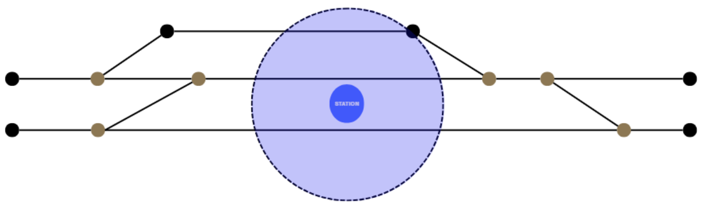

Figure 4.2: Example of platform detection searching zone

This assumption is based on the observation that station platforms are generally located alongside the tracks that serve passenger operations. Therefore, the nearest segments constitute a reasonable approximation of the actual platform locations while requiring only publicly available infrastructure data.

Within the research zone, the script identifies all track segments from the track dataset that intersect with the zone. Then, the script ranks the track segments based on their proximity to the station coordinates. The remaining track segments are classified into three categories (given a station that has n platforms):

1. Best candidates: the track segments that are closest to the station coordinates and are likely to correspond to a platform ($rank \leq n$).

2. Mediocre candidates: the track segments that are within the research zone (or close to it) but are not the best candidate. These segments may correspond to additional platforms or other track segments near the station ($n \leq rank \leq 1.5n$).

3. Irrelevant segments: the track segments that are outside the research zone or are too far from the station coordinates. These segments are unlikely to correspond to platforms and can be disregarded in the platform detection process ($1.5n \leq rank \leq 2n$).

This ranking approach allows the script to prioritize track segments that are most likely to correspond to platforms while minimizing the inclusion of irrelevant segments. By categorizing the track segments based on their proximity to the station coordinates, the script can effectively identify potential platform locations and improve the accuracy of the platform detection process.

The resulting associations are evaluated using distance-based quality indicators. Stations for which the selected segments are located within a short distance are considered to have a high-quality match, whereas larger distances indicate potentially less reliable assignments. This quality assessment is later used to analyse the overall accuracy of the generated infrastructure model.

Track Segmentation

The platform matching process may produce situations in which several stations are associated with the same track segment. Such conflicts occur when a single segment represents

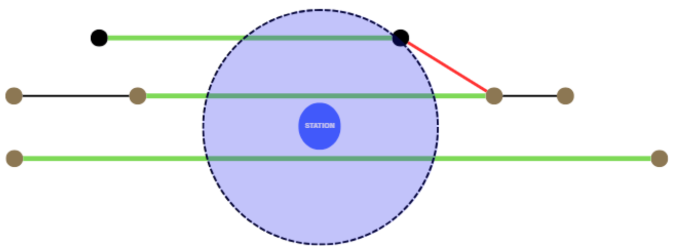

Figure 4.3: Example of platform detection ranking when the number of platform is 3

a long railway section passing through multiple stations, making it impossible to uniquely associate each platform with a distinct operational element.

To resolve this issue, an additional track segmentation procedure is applied after the platform matching stage. The objective is to increase its operational granularity to ensure that each station platform can be uniquely associated with a distinct track segment.

First, all segments assigned to more than one station are identified. Only these conflicting segments are processed, while the remaining network remains unchanged. Each selected segment is then divided into smaller sub-segments according to a maximum segment length threshold of 800 meters.

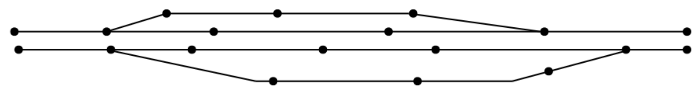

Figure 4.4: Track segmentation Initial setup

The segmentation process follows the existing geometry of the track. New sub-segments are created exclusively using the coordinates already present in the original polyline representation. No interpolation or artificial points are introduced. Distances are computed after projecting the geographical coordinates into the Belgian Lambert 72 coordinate system (EPSG:31370), allowing accurate metric measurements.

For each generated sub-segment, new extremity nodes are created when necessary and the corresponding track attributes are recomputed. The resulting infrastructure therefore preserves the original network geometry while providing a finer representation of the railway infrastructure.

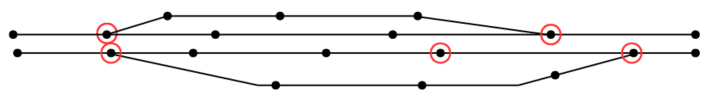

Figure 4.5: Track segmentation applied to a conflicting segment

After segmentation, the platform matching procedure can be repeated on the newly generated segments. The process continues until each station platform is associated with a distinct segment. In practice, a single segmentation iteration was sufficient to eliminate all observed conflicts in the Belgian railway network dataset.

This approach allows the model to maintain a realistic infrastructure representation while ensuring that stations and platforms can be uniquely mapped to simulation entities.

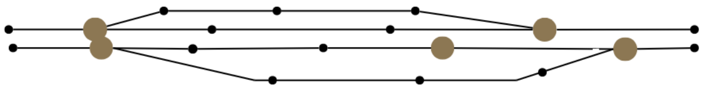

Figure 4.6: Track segmentation results

Such a requirement is particularly important for subsequent routing, timetable generation, and railway traffic simulation tasks.

Dual Graph Construction

Once the railway infrastructure has been segmented into operational track sections, a graph representation is constructed to efficiently compute train routes through the network. The proposed approach adopts a dual graph representation, where each railway track segment becomes a graph node, while the possible transitions between consecutive track segments are represented as directed edges.

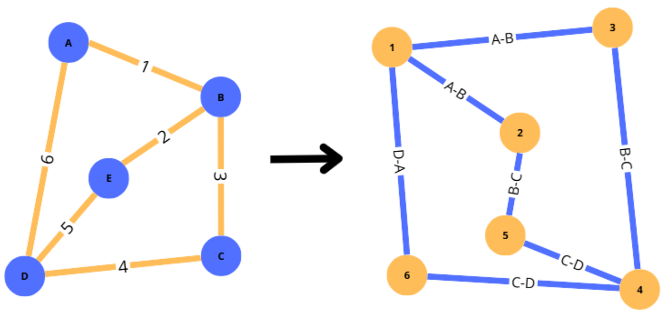

Figure 4.7: Dual graph

This representation considerably simplifies routing operations. Since trains always travel along track sections and move from one section to another through switches or junctions, representing track segments as graph nodes naturally models train movements. Furthermore, routing algorithms operate directly on operational entities instead of geometric infrastructure elements.

The construction of the dual graph is performed in two stages.

First, each track segment is associated with its start and end junctions. A temporary lookup structure is generated in order to quickly retrieve all tracks connected to a given infrastructure node. This preprocessing step accelerates the graph construction by avoiding repeated searches through the complete infrastructure dataset.

Next, the directed graph is created using the `NetworkX` library. For every pair of track segments, a directed edge is added whenever the destination node of the first track corresponds to the origin node of the second track. Formally, given two track segments (t_i) and (t_j), a directed edge is created if

$$\mathrm{end}(t_i) = \mathrm{start}(t_j). \tag{4.1}$$

The resulting edge represents a valid train movement from one track segment to the next. Each edge is assigned a weight corresponding to the physical length of the originating track segment. Consequently, shortest-path computations minimize the travelled distance across the railway infrastructure.

The resulting graph can be formally defined as $G = (V, E)$,where each vertex $(v \in V)$ represents a railway track segment and each directed edge $((u, v) \in E)$ denotes a feasible transition between two consecutive track segments.

Once the dual graph has been generated, shortest paths between stations can be efficiently computed. Since a station may contain multiple platforms, routing cannot simply be performed between station identifiers. Instead, every platform of the origin station is paired with every platform of the destination station. For each platform pair, the corresponding track segments are identified and Dijkstra's shortest-path algorithm, provided by NetworkX, is executed on the dual graph using the track lengths as edge weights.

The algorithm returns the ordered sequence of track segments connecting the two platforms. This sequence is then converted into the corresponding SUMO edge identifiers by identifying the physical connections between consecutive track segments. The resulting edge sequence can be directly inserted into SUMO route definitions without requiring any additional processing.

To avoid repeatedly solving identical routing problems, all computed routes are stored in a cache indexed by the origin station, destination station, and platform pair. Subsequent requests for the same station pair immediately retrieve the previously computed route, significantly reducing the computational cost during trip generation. Finally, whenever the operational dataset contains trips between station pairs that have not yet been processed, the corresponding routes are computed on demand and added to the cache. This strategy guarantees that every station-to-station movement required by the timetable is associated with a valid infrastructure path before the simulation is generated.

4.2.3 Trip Generation

The next step consists of generating train trips that will be executed during the simulation. A trip defines the movement of a train through the railway network by specifying its origin, destination, departure time, and the route computed between these locations. The quality and realism of the generated trips directly influence the representativeness of the simulation and, consequently, the validity of the obtained results.

Different strategies can be adopted for trip generation. Synthetic trips are useful for validating the generated infrastructure, evaluating routing capabilities, and creating controlled traffic scenarios with varying levels of congestion. Conversely, trips reconstructed from real operational data allow the simulation to reproduce actual railway operations and provide a more realistic representation of traffic conditions.

To address both requirements, this work implements two complementary trip generation approaches. The first approach automatically generates random trips over the reconstructed railway network, while the second reconstructs train movements from real operational datasets. The following sections present these two methodologies in detail.

Random Trips

The first trip generation strategy aims at producing synthetic railway traffic over the reconstructed network without relying on historical operational data. This approach is primarily intended to validate the generated infrastructure, verify the connectivity of the network, and evaluate the routing capabilities of the simulation under controlled conditions.

To achieve this, the reconstructed train paths obtained during the infrastructure generation phase are first analyzed in order to identify every station that appears either as the starting point or the terminal point of an existing railway service. These stations constitute the candidate locations from which synthetic trips can be generated.

Once the candidate stations have been identified, weighting files are automatically generated for SUMO's randomTrips.py utility. Each platform associated with a candidate station receives an identical weight, while all other tracks are excluded from the selection process. Consequently, the trip generator only selects railway platforms corresponding to valid departure and arrival stations, preventing the creation of unrealistic trips originating from intermediate tracks, switches, or maintenance sections.

The randomTrips.py tool is then executed on the generated railway network. The generation process is constrained to railway edges only, ensuring that all produced trips are compatible with the rail infrastructure. Several parameters are used to increase the realism of the generated traffic. In particular, a minimum travel distance is imposed between the origin and destination to avoid trivial journeys, while a variable insertion rate is specified throughout the simulation period to emulate fluctuations in railway traffic intensity during the day. The resulting output is a set of synthetic trips, each containing a departure edge, an arrival edge, and an associated departure time.

Although these trips are syntactically valid, they do not necessarily correspond to railway services that can actually be operated within the Belgian railway network. To ensure operational consistency, an additional validation step is performed. For each generated trip, the departure and arrival platforms are first mapped back to their corresponding stations. The algorithm then verifies whether these two stations belong to at least one reconstructed train path extracted from the operational dataset. Only trips whose origin and destination are connected by an existing railway service are retained. Trips failing this verification are discarded.

Finally, the validated trips are converted into a SUMO route file using the duarouter routing engine. This tool computes the complete railway path through the generated infrastructure while respecting the network topology and railway connectivity constraints. The resulting routes are subsequently used as input for the simulation.

Real Data Trips

In addition to synthetic traffic generation, a second approach was developed to reproduce actual railway operations using real punctuality data. Unlike the previous method, which generates artificial train movements, this approach aims at recreating real train services by preserving their scheduled sequence of stops and planned timetable. The objective is to obtain a simulation that closely reflects real railway traffic and therefore provides a more realistic digital twin of the Belgian railway network.

In principle, this approach would allow all train journeys present in the selected operational dataset to be inserted into the simulation. For each train and each operating day, the timetable is reconstructed as an ordered list of station stops. Each station is then associated with one of its detected platforms, and the route between consecutive stations is computed using the dual graph constructed during the infrastructure modeling phase. The final route of a train is obtained by concatenating all the paths computed between successive stops.

The first step consists in processing the punctuality dataset in order to reconstruct each train service individually. The operational records are first cleaned by removing incomplete entries and converting the planned arrival and departure timestamps into simulation time expressed in seconds from the beginning of the simulation. The records are then grouped according to the train identifier and operating date, allowing each train journey to be reconstructed as an ordered sequence of stations associated with their planned arrival and departure times.

Once the timetable has been reconstructed, each station visited by a train is associated with the set of platforms detected during the infrastructure modeling phase. Platform assignment is then performed while respecting the timetable constraints and ensuring that no two trains occupy the same platform simultaneously. A configurable safety buffer is introduced between successive platform occupations in order to prevent unrealistic conflicts and better represent operational constraints encountered in real railway systems.

The main difference with the random trip generation strategy lies in the route computation. Instead of relying on SUMO to determine a shortest path between two stations, this methodology exploits the dual graph introduced in Section 4.2.2. For every pair of consecutive stations in a train service, the corresponding path is searched within the dual graph, taking into account the selected departure and arrival platforms. The complete train route is then obtained by concatenating the paths computed between every pair of successive stations. This approach guarantees that trains follow the detailed railway topology reconstructed during the infrastructure modeling stage and remain consistent with the operational characteristics of the network.

However, although the methodology is conceptually sound, the current implementation did not produce exploitable simulation results. During the platform assignment process, many consecutive station pairs could not be connected because no valid path was found in the generated dual graph. As a consequence, a large proportion of train services were rejected before route generation, preventing the construction of a complete operational timetable.

The absence of feasible paths does not originate from the timetable itself but rather from the current state of the generated dual graph. Some infrastructure elements or connections required to guarantee complete network connectivity are still missing or incorrectly represented, making certain station-to-station journeys impossible despite their existence in the real railway network. Improving the connectivity and completeness of the dual graph therefore constitutes a necessary step before realistic timetable-based simulations can be successfully performed.

Despite these limitations, this approach establishes the foundations for integrating real operational data into the proposed railway digital twin. Once the dual graph reconstruction is sufficiently complete, the methodology will enable the simulation of real train movements while preserving both the actual timetable and the detailed infrastructure topology.

4.2.4 Simulation

Once the micro-level infrastructure has been generated and train trips have been created, the resulting scenario can be simulated in SUMO. In contrast with the macro-level model, the micro-level simulation relies on a detailed representation of the railway infrastructure. Tracks, switches, platforms, and operational segments are explicitly represented, which allows trains to circulate through a much more realistic network topology.

The simulation procedure is designed to support two different types of traffic scenarios. The first scenario uses randomly generated trips, while the second attempts to reproduce real train movements extracted from operational data. These two approaches serve different purposes. Random trips are mainly used to validate the generated infrastructure and to test whether trains can move through the network under controlled conditions. Real-data trips, on the other hand, aim to reproduce actual railway operations and therefore provide a more realistic simulation of the Belgian railway network.

Random-trip simulation

The random-trip simulation is based on synthetic train movements generated from the reconstructed railway network. Although the generated trips do not correspond to real train services, the generation process is constrained in order to make them as realistic as possible. The objective is not to reproduce a specific timetable, but to create plausible railway movements that can be executed on the micro-level infrastructure.

Additional parameters provided by SUMO's trip generation and routing tools are then adjusted to improve the realism of the scenario. The insertion rate controls the frequency at which trains are introduced into the simulation. By modifying this parameter, it is possible to generate different traffic densities and to emulate periods with more or fewer train movements. Other parameters, such as the minimum trip distance, the departure time distribution, and the selection weights assigned to candidate edges, are also used to avoid unrealistic routes and to favour meaningful station-to-station movements.

After the random trips have been generated, they are routed through the micro-level network. The routing step computes complete paths between the selected departure and arrival platforms while respecting the topology of the SUMO network. The resulting routes are then inserted into the simulation configuration. This process makes it possible to test whether the reconstructed infrastructure is sufficiently connected and whether the generated railway assets can support train circulation.

The number of simulated days is configurable by the user, as in the macro-level model. This parameter controls the simulation horizon and influences the number of generated trips. A short duration is useful for validating the network and detecting routing errors, while a longer duration can be used to evaluate the behaviour of the infrastructure under more sustained traffic conditions.

The random-trip simulation therefore plays an important validation role. Since the generated trips are not constrained by a real timetable, they provide a flexible way to test the robustness of the network, the correctness of the platform detection process, the quality of the routing, and the ability of SUMO to execute train movements on the reconstructed infrastructure.

Real-data simulation

The second simulation scenario aims to use real operational data instead of synthetic trips. In this case, the objective is to reproduce the train services contained in the punctuality dataset. Each train movement should be reconstructed from the real sequence of stations visited by the train, associated with the corresponding planned arrival and departure times, and then mapped onto the detailed micro-level infrastructure.

However, the current implementation of this scenario did not produce exploitable simulation results. The main difficulty appears during the mapping of real train movements onto the reconstructed micro-level infrastructure. For majority of station pairs, no valid path could be found in the dual graph. As a consequence, a significant number of train services were rejected before being converted into SUMO routes, preventing the generation of a complete real-data simulation scenario.

This issue does not necessarily indicate that the operational data is incorrect. Rather, it highlights limitations in the current state of the reconstructed micro-level graph. Some connections required to reproduce real railway movements are missing, incorrectly oriented, or not represented with sufficient topological consistency. Since the real-data simulation requires complete paths between all consecutive stations of each train journey, even small connectivity gaps can prevent an entire service from being simulated.

Despite this limitation, the real-data simulation workflow remains an important contribution of the implementation. It defines the structure required to integrate operational railway data into the digital twin and identifies the main technical obstacle that must be solved in future work. Improving the completeness of the dual graph, refining the representation of switches and track directions, and strengthening the platform assignment strategy should make it possible to execute real timetable-based simulations in a future version of the system.

4.2.5 Limitations

The micro-level model offers a more detailed representation of the railway infrastructure, but this additional realism also introduces several limitations. The quality of the simulation strongly depends on the completeness and correctness of the reconstructed infrastructure graph. Missing connections, incorrect track orientations, imperfect switch detection, or inaccurate platform assignments can prevent the computation of valid routes and may lead to unrealistic or incomplete simulations.

A second limitation concerns the integration of real operational data. While random trips can be generated and simulated more easily, the use of real train trajectories requires a continuous and valid path between every pair of consecutive stations in the timetable. In the current implementation, this requirement could not always be satisfied, which prevented the generation of a fully exploitable real-data simulation. These limitations are only introduced here briefly, as they are discussed in more detail in the dedicated limitations chapter.

Chapter 5

Results

This chapter presents the results obtained from the modeling and simulation pipeline developed in this thesis. The evaluation focuses on two main aspects. First, the generated datasets are compared with the available reference data in order to assess their completeness. Second, the infrastructure elements extracted during the modeling process are analysed to evaluate the quality of the generated railway representation.

The results are presented separately for the macro-level and micro-level approaches when necessary. This distinction is important because the two models do not represent the railway network at the same level of detail.

5.1 Dataset Validation

5.1.1 Station Coverage

Station coverage evaluates the number of stations detected and integrated into the generated datasets. This metric is important because stations are central elements in both modeling approaches. In the macro-level model, stations are the main nodes of the graph. In the micro-level model, stations are used to associate platforms and tracks with operational stopping points.

Simulation Level	Generated Stations	Reference Stations	Coverage (%)
Macro-Level	559	559	100
Micro-Level	553	554	99.82

Table 5.1: Station coverage for macro-level and micro-level simulations

The results show that station coverage is very high for both modeling approaches. For the macro-level simulation, all 559 stations present in the reference dataset are generated, corresponding to a coverage of 100.00%. This indicates that, at the station-to-station level, the macro-level generation process is able to fully preserve the set of stations required for the generated network.

For the micro-level simulation, 553 stations are generated from a reference dataset containing 554 stations, corresponding to a coverage of 99.82%. This result is also very strong, especially considering that the micro-level model requires a more detailed association between stations, platforms, and physical track infrastructure. The small difference between the reference and generated datasets may be explained by data inconsistencies, preprocessing filters, or the absence of a valid platform-track association for one station.

Overall, the station coverage results confirm that the generated datasets provide a reliable station basis for the rest of the modeling pipeline. The macro-level approach achieves complete station coverage, while the micro-level approach remains almost complete despite the additional infrastructure constraints introduced by the detailed model.

5.1.2 Platform and Track Coverage

The second part of the dataset validation concerns platform and track coverage. These elements are especially important for the micro-level model, where the objective is to represent the railway infrastructure in more detail. Platforms define where trains can stop inside stations, while tracks define the physical railway segments used for circulation.

Category	Reference Data	Simulation Data	Coverage (%)
Platforms	1551 platforms	1560 platforms	99.42
Tracks (Macro-Level)	1385 tracks	1560 tracks	100
Tracks (Micro-Level)	25028 tracks	26721 tracks	106.59

Table 5.2: Platform and track coverage for micro-level simulation

The platform coverage obtained for the micro-level model is also very high. The reference dataset contains 1560 platforms, while the generated dataset contains 1551 platforms, resulting in a coverage of 99.42%. This shows that the platform detection and matching process is able to identify almost all platforms available in the reference data. However, it does not guarantee that the generated platforms are correctly positioned on the tracks, which is a separate aspect of the infrastructure modeling process.

Track coverage differs between the macro-level and micro-level models. In the macro-level simulation, the reference data contains 1385 station-to-station tracks and the generated dataset also contains 1385 tracks, giving a coverage of 100%. This result is expected because the macro-level model directly represents connections between stations and does not attempt to reconstruct the detailed internal railway topology.

For the micro-level simulation, the generated dataset contains 26721 tracks, while the reference dataset contains 25068 tracks. This corresponds to a coverage of 106.59%. A value higher than 100% does not necessarily indicate an error. It reflects the fact that the micro-level generation process may create more detailed or segmented track representations than those present in the reference dataset. In particular, a single reference track can be represented by multiple generated segments after preprocessing, switch detection, and track segmentation.

Overall, the platform and track coverage results show that the generated infrastructure is highly complete. However, the interpretation of track coverage must take into account the difference between reference tracks and simulation-ready SUMO edges. The generated micro-level network is not only a copy of the reference data, but a transformed representation adapted to routing and simulation constraints.

5.2 Infrastructure Modeling Results

After validating the general completeness of the datasets, the next step is to evaluate the results of the infrastructure modeling process. This part focuses on the railway elements that were extracted, transformed, or reconstructed during the implementation. The objective is to assess whether the generated infrastructure provides a coherent basis for railway simulation.

The infrastructure modeling results confirm the high level of coverage already observed in the dataset validation step. Stations are almost completely detected in both modeling approaches, and platforms are also recovered with a high success rate in the micro-level model. Switches require a more nuanced interpretation, because they are inferred from the generated railway topology rather than directly copied from the reference dataset.

Category	Reference Data	Simulation Data	Detection (%)
Stations (Macro-Level)	559	559	100
Stations (Micro-Level)	554	553	99.82
Platforms (Micro-Level)	1560	1551	99.42
Switches (Micro-Level)	10379	12343	118.92

Table 5.3: Detected stations, platforms and switches for macro-level and micro-level simulations

The detection percentage reaches 100.00% for macro-level stations and 99.82% for micro-level stations. Platform detection reaches 99.42%, which confirms that most platforms could be matched to the generated infrastructure. These results are important because platforms are used as stopping locations for trains and as origin or destination points during trip generation.

Switch detection provides a different type of result. The reference data contains 10379 switches, while the generated micro-level infrastructure contains 12343 detected switches, corresponding to a coverage of 118.92%. As with micro-level track coverage, this value above 100% must be interpreted carefully. It does not simply mean that the model detects more correct switches than expected. Instead, it indicates that the algorithm identifies more candidate switch points than the number of switches available in the reference dataset.

This visual comparison makes it possible to understand how the algorithm enriches the infrastructure by identifying branching and merging points across the network.

This difference can be explained by the way switches are inferred from the railway topology. In the generated network, a switch is detected when a node connects more than two track segments and therefore behaves as a branching or merging point. However, this topological definition may identify some elements that are not classified as switches in the reference data. It may also split complex junctions into several smaller switch-like elements.

The visual comparison of the network before and after switch detection helps illustrate this effect. The switch detection process enriches the generated infrastructure by identifying relevant branching points and making the network more suitable for routing. In dense areas such as Bruxelles-Midi, the detected switches provide a much more detailed representation of the railway topology, but they also highlight the difficulty of matching generated topological elements exactly with reference infrastructure objects.

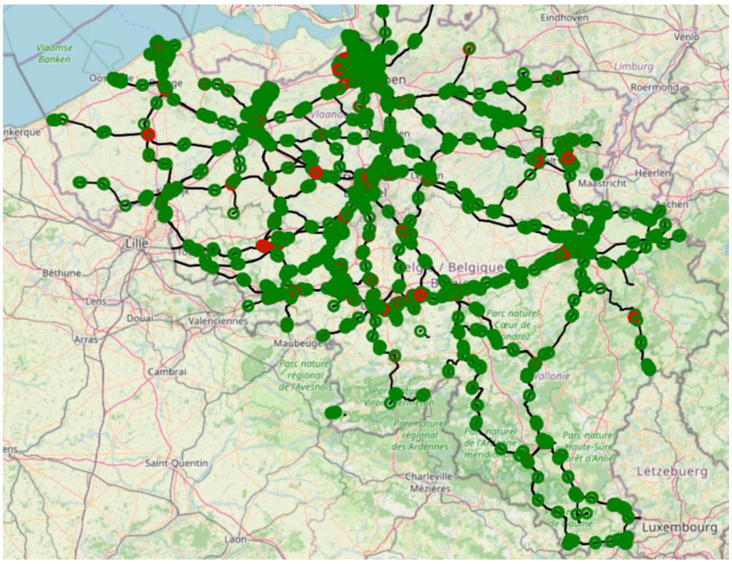

Figure 5.1: Network before and after switch detection (Green dots represent detected switches with 3 connections and Red dots represent detected switches with 4 or more connections)

This station is particularly relevant because it contains a dense set of tracks and platforms, making it a useful example for evaluating whether the platform matching algorithm can handle complex station environments.

The platform matching figures further illustrate the quality of the generated infrastructure. They show how platforms are associated with the corresponding railway tracks and make it possible to visually inspect the quality of the matching process. This qualitative validation complements the numerical coverage metrics and helps identify local cases where the matching may be less accurate.

The platform matching quality is further illustrated by a qualitative comparison of several representative stations. As shown in Figure 5.4, the matching process can produce different levels of quality depending on the local infrastructure configuration. For instance, Tubize is classified as a good-quality match, with a very small minimum distance between the detected platform and the candidate track, which suggests a reliable association. Weerde corresponds to an intermediate case, labelled as a warning, where the matching remains acceptable but presents a larger average distance, indicating a less precise association. By contrast, Hansbeke is classified as insufficient quality, showing that the algorithm may encounter difficulties in some local configurations, especially when the search radius is larger or when the number of candidate tracks does not allow a clear correspondence to be established.

5.3 Simulation Results

The previous sections evaluated the completeness of the generated datasets and the quality of the reconstructed infrastructure. This section focuses on the simulation results obtained from the two implemented modeling approaches. As in the implementation chapter, the results are separated between the macro-level and micro-level simulations because the two models do not have the same objective or the same level of detail.

The macro-level simulation is evaluated through a real-data scenario executed on a

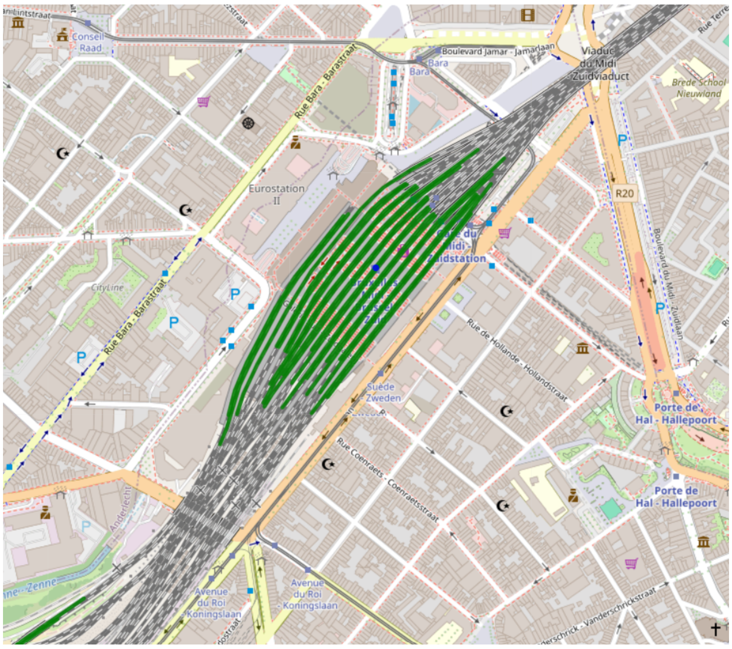

Figure 5.2: Platform matching result on Bruxelles-Midi

small chain of neighbouring stations. The micro-level simulation is evaluated through a random-trip scenario executed on the reconstructed detailed network. These two results illustrate the current simulation capabilities of the proposed framework and highlight the different roles of the two levels of representation.

5.3.1 Macro-Level Simulation Results

The macro-level simulation was successfully executed on a selected chain of four neighbouring stations: Cour-sur-Heure, Ham-sur-Heure, Berzée, and Pry. This scenario was chosen because it corresponds to a limited and coherent part of the railway network, with moderate traffic intensity and a simple station-to-station structure.

The simulation uses real operational data extracted from the punctuality dataset. Train movements are reconstructed from the observed sequence of stations and converted into SUMO routes. This means that the simulated traffic is not randomly generated: it corresponds to real train services that circulated through the selected part of the network during the considered time window. The result therefore validates the ability of the macro-level pipeline to connect infrastructure data, operational data, route generation, and SUMO simulation execution.

Element	Value
Simulated Area	Cour-sur-Heure, Ham-sur-Heure, Berzée, Pry
Number of Stations	4
Trip generation method	Real operational data
Simulation duration	5 days
Number of generated train routes	115
Number of complete simulated trips	115

Table 5.4: Macro-level simulation scenario summary

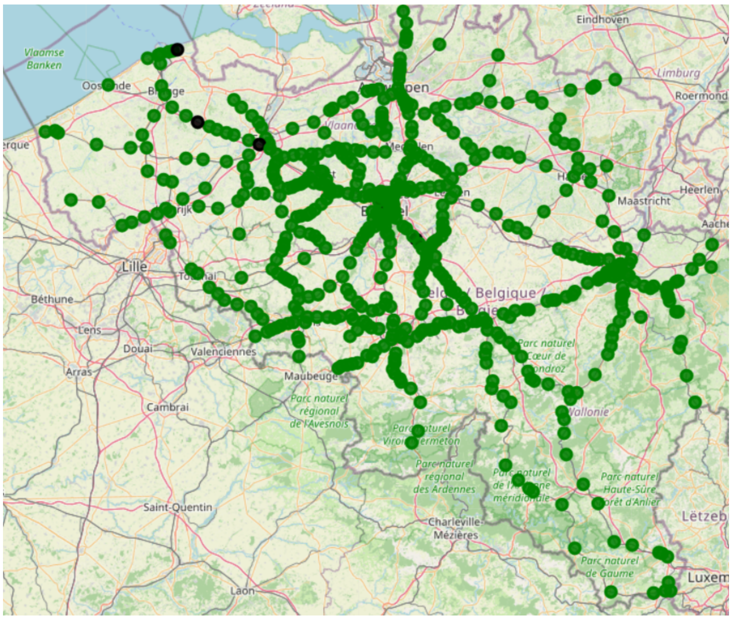

Figure 5.3: Platform matching quality map

Figure 5.4: Platform matching quality examples

The main result of this scenario is that the macro-level approach is able to execute a simulation based on real train movements, provided that the selected part of the network remains compatible with the assumptions of the model. In particular, the approach works best on simple chains of neighbouring stations where each station has a limited local complexity. Stations with two main directions are especially suitable because the simplified station-node representation remains close to the actual operational structure. In contrast, dense stations with multiple platforms, branching directions, and high traffic intensity are less appropriate for this abstraction because internal routing and platform constraints are not represented.

This result confirms the usefulness of the macro-level model as a first validation layer for the proposed DTR. It demonstrates that real operational data can be transformed into executable SUMO routes when the selected network area is sufficiently simple and coherent. However, the result should not be interpreted as a complete validation of real railway operations. The macro-level simulation validates station-to-station circulation, but it does not validate detailed infrastructure behaviour inside stations.

5.3.2 Micro-Level Simulation Results

The micro-level simulation was evaluated using randomly generated trips over the reconstructed detailed railway network. As explained in Chapter 4, the real-data micro-level scenario could not be fully executed because the mapping of real train movements onto the

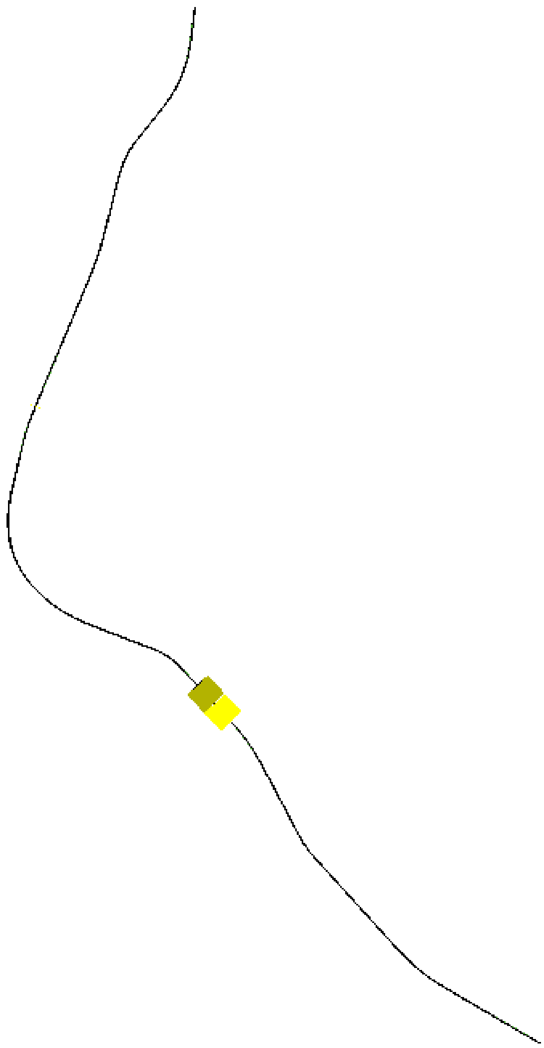

Figure 5.5: Macro-level simulation network for the selected four-station chain

detailed infrastructure graph did not provide valid paths for a sufficient number of station pairs. For this reason, the result presented here focuses on the random-trip scenario.

The random-trip simulation was executed on the entire reconstructed micro-level network. Trips were generated between valid platform locations and then routed through the SUMO network. The objective of this scenario is therefore not to reproduce a real timetable, but to evaluate whether the generated infrastructure can support train circulation at the micro level.

The results show that 16802 train routes were generated and successfully routed. This is an important result because it confirms that the generated infrastructure is not only a static representation of railway tracks, but can also be used by SUMO to compute executable train movements. In this sense, the random-trip simulation validates the connectivity of the generated network and demonstrates that the reconstructed micro-level infrastructure is usable for simulation purposes.

However, the number of completed trips is significantly lower than the number of generated and routed trips. Out of 16802 generated routes, 4912 trips were completed during the simulation. This difference must be interpreted carefully. It does not necessarily indicate that the generated network is disconnected, since the routes were successfully

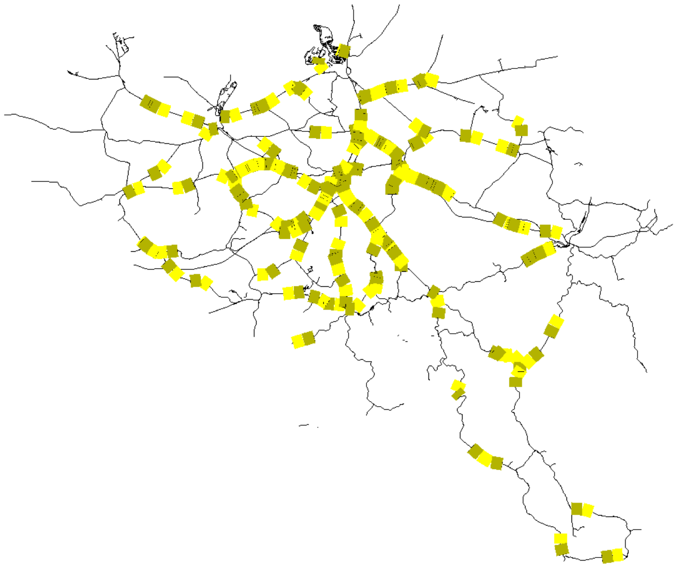

Figure 5.6: SUMO-GUI screenshot of the micro-level random trip simulation

Element	Value
Simulated Area	Entire reconstructed micro-level network
Number of Stations	553
Number of Platforms	1551
Number of Tracks	26721
Trip generation method	Random trips between valid platforms
Simulation duration	1 day
Number of generated train routes	16802
Number of successfully routed trips	16802
Number of completed trips	4912

Table 5.5: Micro-level simulation scenario summary

computed. Rather, it reflects the limitations of random traffic generation when applied to a detailed railway network.

Random trips are generated without reproducing the full set of operational constraints that would normally be considered in a real railway timetable. In particular, the generation process does not fully account for train speeds, headways, platform occupation, route conflicts, junction capacity, or the temporal separation between incompatible movements. As a result, many trains may be inserted too close to one another or may attempt to use conflicting parts of the infrastructure at similar times. During the simulation, this can lead to warnings such as emergency braking, collisions, teleportation events, or blocked insertion areas.

```
Warning: Block after insertion lane '29646_0' exceeds maximum length
(stopped searching after edge ':2794_0' (length=20207.42m)).
Warning: Vehicle '12897' performs emergency braking on lane '49375_0'
with decel=5.00, wished=1.00, severity=1.00, time=85488.00.
Warning: Teleporting vehicle '12897'; collision with vehicle '10232',
lane='49527_0', gap=-40.28, time=85489.00, stage=move.
```

These warnings are therefore consistent with the nature of the generated scenario. They show that the infrastructure can be used for routing, but that random trip generation does not provide a realistic operational timetable. The simulation should consequently be interpreted as a technical validation of the generated network rather than as a realistic reproduction of railway operations. It demonstrates that trains can circulate on the reconstructed micro-level infrastructure, while also showing that a more constrained and timetable-aware trip generation method is required to obtain operationally meaningful results.

Overall, the micro-level simulation result confirms the feasibility of executing train movements on the detailed network. The fact that all generated trips could be routed suggests that the network contains a high level of connectivity from the point of view of SUMO routing. However, the lower number of completed trips highlights that routing feasibility alone is not sufficient to produce realistic railway simulation. Future improvements should therefore focus not only on infrastructure connectivity, but also on the generation of traffic scenarios that better account for railway operational constraints.

5.4 Computational Performance

In addition to evaluating the quality of the generated railway infrastructure and the execution of simulation scenarios, it is also important to analyse the computational performance of the proposed pipeline. This evaluation provides insight into the cost of transforming the initial railway datasets into simulation-ready SUMO configuration files. It also helps identify which steps of the implementation are the most time-consuming and where future optimizations should be prioritized.

The performance analysis is divided into two complementary parts. The first part focuses on the generation pipeline, from the initial ETL operations to the creation of the network, route, stop, and configuration files required by SUMO. The second part focuses on the execution of the simulation itself, with particular attention to the micro-level model, which contains a more detailed infrastructure representation and is therefore more computationally demanding.

5.4.1 Pipeline Generation Performance

The first evaluation concerns the execution time of the main processing steps required to generate a simulation scenario. For both the macro-level and micro-level models, the pipeline can be decomposed into several stages: data loading and preprocessing, infrastructure generation, route or trip generation, additional file generation, and final simulation configuration.

The longer execution time of the micro-level pipeline is therefore expected. It is mainly caused by the larger number of infrastructure elements and by the additional processing stages required to construct a simulation-ready detailed network. In particular, the micro-level pipeline includes switch detection, platform detection, track segmentation, detailed network generation, and random trip generation. These operations are not required, or are much simpler, in the macro-level model.

The comparison between the macro-level and micro-level generation pipelines shows a clear difference in computational cost. The macro-level pipeline completes in approximately 331 seconds, while the micro-level pipeline requires approximately 775 seconds.

Pipeline Step	Execution Time (s)
ETL	330.92
Network generation	0.12
Trip generation	15.08
Additional file generation	0.0058
Configuration generation	0.0000
Total	331.0458

Table 5.6: Execution time of the macro-level generation pipeline for the example network

Pipeline Step	Execution Time (s)
ETL	344.09
Switch detection	1.50
Platform Detection	19.07
Track segementation	81.90
Network generation	160.37
Trip generation	168.30
Additional file generation	0.02
Configuration generation	0.00
Total	775.25

Table 5.7: Execution time of the micro-level generation pipeline with random trip generation

This result is coherent with the structure of the two models. The macro-level model manipulates a simplified station-to-station graph, whereas the micro-level model processes a much larger and more detailed infrastructure representation.

Among the micro-level processing steps, network generation and trip generation are particularly costly. This is consistent with the fact that the micro-level network contains many more track segments and requires more complex route computation than the macro-level model. Track segmentation also introduces a non-negligible cost, since it modifies the structure of the infrastructure in order to increase the operational granularity of the network.

This result highlights the trade-off between abstraction and realism. The macro-level model is faster and more robust for real-data simulation on selected station chains, but it cannot represent detailed infrastructure constraints. The micro-level model is more expensive to generate, but it provides a richer infrastructure representation that can support more detailed simulations. The additional computational cost is therefore coherent with the higher level of detail introduced by the micro-level approach.

5.5 Discussion

The results presented in this chapter show that the proposed pipeline is able to generate and simulate railway networks at two complementary levels of representation. The macro-level model provides a simplified but robust framework for integrating real operational data into SUMO. The micro-level model provides a more detailed representation of the physical railway infrastructure and demonstrates that train circulation can be simulated on a network composed of tracks, switches, platforms, and segmented railway sections.

The dataset validation results indicate that the generated infrastructure is highly complete. Station coverage is almost complete for both modeling levels, and platform coverage is also very high. Track coverage must be interpreted more carefully, especially for the micro-level model, where the number of generated tracks exceeds the number of reference tracks. This does not necessarily indicate an error. It reflects the fact that the generated SUMO network is a transformed and more segmented representation of the original infrastructure data. The objective is not only to copy the reference dataset, but to create a simulation-ready representation that can be used for routing and train movement simulation.

The infrastructure modeling results also show the value of the micro-level approach. Switch detection and platform matching make it possible to enrich the raw infrastructure data and transform it into operational simulation artefacts. The platform matching quality examples show that the matching process can produce different levels of reliability depending on the local configuration of the station. Some stations provide clear and accurate matches, while others require larger search radii or present more ambiguous candidate tracks. This confirms that the method is effective overall, but that local infrastructure complexity remains an important factor.

The simulation results highlight the complementarity between the two modeling levels. At the macro level, real operational data can be used directly on selected station chains, provided that the selected area remains compatible with the simplifying assumptions of the model. This confirms the usefulness of the macro-level approach as a first end-to-end validation of the digital twin workflow. It demonstrates that infrastructure data, punctuality data, route generation, simulation execution, and output processing can be connected in a coherent pipeline.

At the micro level, the results are different but equally informative. The random-trip simulation confirms that the detailed network can be used by SUMO to route and execute train movements. The fact that 16802 trips were generated and successfully routed suggests that the reconstructed infrastructure is connected from a routing perspective. However, only 4912 trips were completed during the simulation. This difference shows that network connectivity alone is not sufficient to guarantee a realistic and stable railway simulation.

The lower number of completed trips is mainly explained by the nature of random trip generation. Random trips are useful for testing network connectivity and simulation feasibility, but they do not reproduce the constraints of a real timetable. They do not fully account for train speeds, headways, route conflicts, junction capacity, platform occupation, or safe temporal separation between trains. As a consequence, the simulation may generate operational conflicts, leading to emergency braking, teleportation events, insertion problems, or collision-related warnings. These events should not be interpreted as a failure of the infrastructure generation alone, but as evidence that random traffic generation is not sufficient to reproduce realistic railway operations.

The computational performance results further support the distinction between the two approaches. The macro-level pipeline is faster because it manipulates fewer entities and relies on a simpler graph structure. The micro-level pipeline requires more time because it processes a larger amount of data and performs additional operations such as switch detection, platform matching, track segmentation, network generation, and route generation. This additional cost is coherent with the increased realism of the model. It reflects a natural trade-off: the macro-level model is more scalable and easier to use with real operational data, while the micro-level model provides a richer infrastructure basis but requires more processing and more careful traffic generation.

These findings also clarify the next challenge for the proposed digital twin. The micro-level network appears sufficiently connected to support random routing, but the real-data routing strategy based on the dual graph did not yet provide exploitable timetable-based simulation results. This suggests that the next difficulty is not only to generate a connected network, but to identify the correct physical paths corresponding to real train movements. The dual graph approach provides a useful methodological foundation, but future work may need to explore alternative routing strategies, improved direction handling, better platform assignment, or precomputed station-to-station path libraries.

Overall, the results show that the proposed framework successfully establishes the foundations of a railway digital twin for simulation. The macro-level model validates the integration of real operational data in a simplified setting, while the micro-level model validates the reconstruction and simulation of a detailed infrastructure network. The current implementation does not yet provide a fully realistic micro-level timetable simulation, but it identifies the main technical obstacles that must be addressed. In this sense, the contribution of the thesis is not limited to the final simulation outputs, it also lies in the structured analysis of the trade-offs between abstraction, realism, computational cost, and operational feasibility.

Chapter 6

Limitations

This chapter discusses the main limitations encountered during the development of the proposed railway digital twin simulation pipeline. Some limitations were already briefly introduced in the implementation chapter, but they are analysed here in more detail. The objective is not only to identify what prevented the system from reaching a complete operational simulation, but also to clarify which issues are related to data availability, modeling assumptions, routing strategies, simulator constraints, and generalization.

The proposed framework successfully demonstrates that railway infrastructure and operational data can be transformed into SUMO-compatible simulation artefacts. However, the results also show that building a realistic railway simulation from heterogeneous real-world datasets remains a difficult task. This is especially visible at the micro level, where the objective is to reproduce the detailed physical structure of the network and to map real train movements onto this infrastructure. The limitations discussed in this chapter therefore provide important insights for future work on railway digital twins.

6.1 Data Availability and Quality

A first important limitation concerns the availability and precision of the infrastructure data. Although the available datasets provide a strong basis for reconstructing the railway network, some information required for a fully realistic simulation is either missing, incomplete, or not available at the level of detail required by the simulator. This issue is particularly important for the micro-level model, where the simulation depends on the accurate representation of tracks, switches, platforms, and their spatial relationships.

One of the clearest examples concerns platform positions. The available data provides information about stations and the number of platforms, but it does not always provide the exact physical position of each platform in the railway infrastructure. As a result, platforms had to be reconstructed using geometric methods, mainly by associating each station platform with nearby track segments. This approach makes it possible to generate platform candidates and to create SUMO train stops, but it also introduces uncertainty. Even when the correct number of platforms is detected, their exact placement may not correspond to the real infrastructure. A platform may be associated with a neighbouring track instead of the operationally correct one, especially in complex stations where several tracks are close to each other.

This limitation has a direct impact on simulation quality. In SUMO, train stops must be attached to specific lanes. If a platform is associated with an incorrect lane, the train may stop at a location that is geographically close to the station but operationally unrealistic. In some cases, the selected track may also be difficult or impossible to reach from the reconstructed route. Therefore, platform detection is not only a data enrichment step, but also a critical dependency for the feasibility of the micro-level simulation.

Another limitation concerns operational route data. The punctuality dataset describes train passages through stations and provides temporal information such as planned and observed arrival or departure times. However, it does not provide the exact physical route followed by each train inside the railway infrastructure. In other words, the data indicates that a train moved from one station to another, but it does not specify which tracks, switches, platforms, or junctions were actually used. This absence of route-level information is one of the main obstacles to constructing a complete micro-level simulation from real data.

Because of this limitation, train paths had to be reconstructed manually or algorithmically from station sequences. This task is considerably more complex at the micro level than at the macro level. Without reference data describing the possible or historical paths used by trains, the reconstruction process must infer these paths from infrastructure topology alone. This makes the result highly dependent on the completeness and correctness of the generated graph.

A dataset describing possible train paths, authorized routes, or historical physical trajectories would significantly improve the pipeline. Such a dataset would provide a direct link between operational records and infrastructure elements. It would also make it possible to distinguish between several possible routes connecting the same pair of stations, which is essential in dense railway areas. Without this information, the simulation can reconstruct plausible paths, but it cannot guarantee that these paths correspond to the routes actually used in railway operations.

6.2 Modeling Assumptions

The proposed methodology relies on two levels of abstraction: a macro-level model and a micro-level model. This distinction is useful because it makes it possible to compare a simple and robust representation with a more detailed but more fragile one. However, each level introduces specific modeling assumptions that limit the realism of the resulting simulation.

At the macro level, stations are represented as single nodes and railway links as direct edges between stations. This abstraction is useful for integrating real operational data, because punctuality records are also expressed mainly at the station level. However, this representation hides the internal structure of stations. Platforms, switches, parallel tracks, junctions, signals, and local routing constraints are not represented explicitly. Consequently, the macro-level model cannot reproduce conflicts occurring inside stations, platform occupation constraints, or detailed route interactions near complex junctions.

The macro-level model is therefore mainly suitable for selected station chains with moderate traffic. It provides a functional end-to-end validation of the pipeline, but it cannot be interpreted as a complete operational simulation of the railway system. This is why the macro-level simulation was restricted to selected parts of the network instead of attempting to reproduce the full national traffic. In dense areas, the internal structure of stations and junctions becomes too important to be ignored.

At the micro level, the opposite limitation appears. The model represents the infrastructure in greater detail, but this increases the sensitivity of the pipeline to small data errors. Missing connections, incorrect directions, ambiguous platform assignments, or imperfect switch detection can prevent valid routes from being found. In a detailed infrastructure graph, a small topological inconsistency may invalidate an entire train journey, because the route generation process requires a continuous path between each pair of consecutive stations.

Another important modeling assumption concerns the use of geometric proximity for platform matching. The method assumes that the closest relevant track segments are likely to correspond to the station platforms. This assumption is often reasonable, but it does not always hold in complex railway environments. Large stations, yards, parallel tracks, underground sections, or curved station layouts can create ambiguous spatial configurations. In such cases, geometric proximity alone may not be sufficient to identify the correct operational platform.

These assumptions show that the micro-level model should be understood as a simulation-ready reconstruction of the railway infrastructure rather than as an exact copy of the real network. The objective was to generate a usable infrastructure model from available data, but the absence of some operational and infrastructure details limits the degree of realism that can be achieved.

6.3 Route Reconstruction and Micro-Level Simulation

Another group of limitations is related to the use of SUMO as the simulation platform. SUMO provides a flexible and scriptable environment for generating networks, defining routes, running simulations, and extracting outputs. This flexibility was one of the main reasons for selecting it in this thesis. However, SUMO was not originally designed as a railway-specific operational planning tool. Some railway-specific concepts therefore require additional modeling effort.

In particular, detailed signalling, interlocking, dispatching, platform assignment, route reservation, and railway capacity management are not automatically reproduced unless they are explicitly encoded in the generated files or controlled through additional logic. This is an important limitation for the micro-level simulation, because detailed infrastructure alone is not sufficient to reproduce realistic railway operations. A network may be physically connected, but the simulation may still generate unrealistic behaviour if operational constraints are not represented.

This issue is visible in the random-trip simulation. The generated trips can be successfully routed through the network, but they are not based on a real timetable and do not fully account for headways, platform occupation, train speeds, junction capacity, or safe temporal separation between incompatible movements. As a consequence, trains may be inserted too close to one another or may attempt to use conflicting infrastructure sections at similar times. This can lead to warnings such as emergency braking, blocked insertion areas, teleportation events, or collision-related messages.

These warnings should not be interpreted only as failures of the generated infrastructure. They also reflect the limitations of unconstrained random trip generation in a detailed railway network. In real railway operations, train movements are planned and regulated to avoid such conflicts. Without an equivalent timetable generation or dispatching layer, the simulator receives routes and departure times that may be technically valid but operationally unrealistic.

A further limitation concerns calibration. The simulation parameters used in SUMO, such as speed, dwell time, insertion rate, routing parameters, and trip generation settings, influence the behaviour of the simulation. In the current work, these parameters were selected to obtain plausible and executable scenarios, but they were not calibrated against a complete operational reference model. As a result, the simulation can be used to validate the feasibility of the pipeline, but not yet to provide precise quantitative estimates of railway performance.

6.4 Simulation Constraints in SUMO

The proposed pipeline also faces limitations related to computational complexity. The macro-level approach is relatively efficient because it manipulates a reduced graph composed mainly of stations and station-to-station links. This makes it easier to generate routes from real operational data and to execute simulations on selected station chains.

The micro-level approach is more demanding. It processes a much larger number of infrastructure elements and applies additional operations such as switch detection, platform matching, track segmentation, network generation, and route computation. This additional cost is expected because the model provides a richer representation of the infrastructure. However, it also limits the ease with which the approach can be scaled to large scenarios with real operational traffic.

The difference between routing feasibility and simulation completion is also important. A route can be computed successfully, but the corresponding train may still fail to complete its trip during simulation because of timing conflicts, blocked lanes, unrealistic insertion times, or interactions with other trains. This means that scalability is not only a question of graph size or computation time. It also depends on the quality of the generated timetable and on the operational consistency of the simulated traffic.

For this reason, future versions of the pipeline would need to include additional mechanisms for timetable validation, route conflict detection, and traffic regulation before simulation execution. These mechanisms would help ensure that the generated scenario is not only connected, but also operationally feasible.

6.5 Scalability and Computational Complexity

The final limitation concerns the generalization of the proposed approach. The pipeline was developed and evaluated using the available Belgian railway datasets and the specific structure of the data used in this thesis. Although the general methodology could be transferred to other railway networks, several components would require adaptation.

First, infrastructure datasets may differ significantly between countries or infrastructure managers. Station identifiers, track geometries, platform descriptions, switch representations, and coordinate systems may follow different conventions. The preprocessing and matching rules developed in this work may therefore not be directly applicable to another network.

Second, operational data may also differ in structure and granularity. Some datasets may contain only planned timetables, while others may include real-time train positions, platform assignments, delay causes, or detailed route information. The quality of the resulting digital twin would depend strongly on the type of operational data available.

Third, the assumptions used in the macro-level and micro-level models may not be equally valid for all railway systems. A simple station-to-station abstraction may be sufficient for low-density regional lines, but less appropriate for dense metropolitan networks. Similarly, a micro-level reconstruction based on geometric proximity may work well in some areas but fail in complex multi-level stations or yards.

The proposed framework should therefore be seen as a methodological foundation rather than a universal solution. It demonstrates how railway infrastructure and operational data can be transformed into simulation-ready artefacts, but it also shows that the quality of the final simulation depends on the availability of detailed data, the correctness of the reconstructed infrastructure, the realism of the routing strategy, and the operational constraints represented in the simulator.

Overall, these limitations do not invalidate the contribution of the thesis. On the contrary, they clarify the main technical barriers that must be addressed to move from a simulation-oriented prototype toward a more complete railway digital twin. The macro-level model validates the integration of real operational data in a simplified setting, while the micro-level model identifies the additional requirements needed for realistic infrastructure-based simulation. Future work should therefore focus on improving platform localization, integrating route-level operational data, replacing single shortest-path routing with multiple feasible path strategies, and adding timetable and conflict-management mechanisms to the simulation pipeline.

Chapter 7

Conclusion

This thesis addressed the construction of a simulation-oriented digital twin for railway network simulation using SUMO and heterogeneous railway datasets. The objective was not to develop a complete operational digital twin, but rather to study the foundations required to transform infrastructure and operational data into simulation-ready railway models.

The work focused on two complementary levels of representation. The macro-level model represents stations as nodes and railway connections as edges, while the micro-level model represents infrastructure elements such as tracks, platforms, switches, and operational segments. This distinction made it possible to evaluate the trade-off between abstraction, scalability, realism, and simulation feasibility.

7.1 Summary of Contributions

The first contribution of this thesis is the design and implementation of a complete data-driven pipeline for railway simulation generation. This pipeline includes data preprocessing, network generation, trip generation, simulation execution, and output analysis. It provides a structured methodology for converting railway datasets into SUMO-compatible simulation artefacts.

The second contribution is the development of a macro-level railway model. This model provides a simplified but functional representation of the railway network and allows real operational data to be integrated into simulation scenarios. It validates the complete workflow from punctuality data processing to SUMO simulation.

The third contribution is the development of a micro-level railway model. This model reconstructs the railway infrastructure in greater detail by representing tracks, switches, platforms, and segmented track sections. It also introduces routing structures, such as the dual graph, that support path computation inside the detailed infrastructure.

Finally, this thesis contributes by identifying the main technical challenges involved in railway digital twin construction. These challenges include data quality, infrastructure completeness, platform matching, route reconstruction, SUMO-specific constraints, and computational scalability.

7.2 Key Findings

The results show that a railway simulation-oriented digital twin can be constructed progressively by combining different levels of abstraction. The macro-level model proved to be more robust for integrating real operational data, since train movements can be represented as station-to-station trajectories. This makes it suitable for validating the overall pipeline and for running simulations on selected parts of the network.

The micro-level model provides a more realistic infrastructure representation, but it also introduces greater complexity. Although the generated infrastructure reaches high coverage and can support synthetic train movements, real-data-based simulation remains difficult. The main issue is that real train trajectories require valid and continuous paths between all consecutive stations, which is not always guaranteed in the reconstructed micro-level graph.

The results also highlight that high dataset coverage does not automatically ensure operational correctness. A station, platform, or track may be detected, but still be insufficiently aligned with the routing requirements of a real railway service. Therefore, railway digital twin construction requires not only good data coverage, but also topological consistency and operational validity.

Overall, the main finding is that macro-level and micro-level models are complementary. The macro-level model is better suited for robust and scalable simulation based on real train movements, while the micro-level model provides a foundation for more detailed infrastructure analysis.

7.3 Practical Implications

This work shows that existing railway data can be transformed into executable simulation models through a structured processing pipeline. This is useful for researchers and railway stakeholders who want to analyse railway operations without manually constructing simulation networks.

The macro-level model can serve as a practical tool for preliminary traffic analysis, selected line simulation, and validation of operational data integration. Its simplified structure makes it easier to use and more robust when detailed infrastructure data is not required.

The micro-level model, although still limited for real-data simulation, provides a basis for more detailed studies of railway infrastructure. With further improvements, it could support analyses related to platforms, switches, local bottlenecks, routing conflicts, and infrastructure capacity.

The work also shows that SUMO can be used as a flexible platform for railway digital twin prototyping. However, railway-specific elements such as signaling, interlocking, dispatching, and precise platform assignment require additional modeling beyond the standard SUMO workflow.

7.4 Future directions

The work presented in this thesis provides a first implementation of a railway digital twin based on real infrastructure and operational data. However, several extensions could be considered in order to transform the current framework into a more complete decision-support tool.

7.4.1 Counterfactual Simulation and What-If Scenarios

A first direction concerns counterfactual simulation and what-if scenarios. The current framework mainly reproduces existing infrastructure and train movements. A future extension could allow users to modify parts of the system and observe the effect of these changes on railway traffic.

For example, it could be possible to simulate the closure of a station, the removal of a track section, a modification of train frequency, or a change in departure times. At the macro level, this could be implemented by modifying station-to-station connections. At the micro level, it could be implemented by making specific platforms, switches, or track segments unavailable.

This would make the digital twin useful as an exploratory tool for testing operational decisions before applying them to the real network.

7.4.2 Tracks Works Planning

A second direction concerns track works planning. Maintenance operations often require temporary closures, reduced capacity, speed restrictions, or rerouting. A simulation-based digital twin could help evaluate the impact of these interventions before they take place.

In the current framework, this could be implemented by adding availability constraints to infrastructure elements. At the macro level, railway connections could be removed or penalized. At the micro level, specific tracks, platforms, or switches could be marked as unavailable during a given time window.

This extension would help compare maintenance strategies, identify bottlenecks, and reduce the impact of planned works on train operations.

7.4.3 Real-Time Forecasting and Incident Propagation

A third direction is the integration of real-time forecasting and incident propagation. The current implementation is mainly based on historical and static datasets. A future version could integrate live or near-real-time information, such as train delays or incidents, and use it to predict the future state of the network.

This could be implemented by combining the simulation pipeline with statistical or machine learning models trained on punctuality data. The macro-level model could be used to study delay propagation between stations, while the micro-level model could help analyse the role of local infrastructure constraints.

Such an extension would make the digital twin useful for operational monitoring and proactive traffic management.

7.4.4 Framework Generalization

A final direction concerns the generalization of the framework. The current implementation is adapted to the Belgian railway datasets used in this thesis. A more general version should separate the core modeling logic from dataset-specific processing steps.

This could be achieved by introducing modular data adapters that transform different railway datasets into a common internal representation. Depending on the available data, the framework could then generate a macro-level model, a micro-level model, or a hybrid representation.

This would increase the reproducibility of the approach and make it easier to apply the methodology to other railway networks.

7.5 Final Remarks

This thesis has shown that constructing a railway digital twin simulation is an incremental process. Before advanced capabilities such as real-time prediction or decision support can be achieved, it is necessary to build reliable simulation models from available infrastructure and operational data.

The proposed macro-level and micro-level approaches demonstrate two complementary ways of addressing this challenge. The macro-level model provides a robust and scalable representation for integrating real train movements, while the micro-level model provides a more detailed infrastructure representation and highlights the challenges of realistic route reconstruction.

The main conclusion of this work is that railway digital twin development requires a balance between simplicity and realism. A simplified model is easier to simulate and more robust, but cannot represent detailed operational constraints. A detailed model is more realistic, but depends more strongly on data completeness and topological correctness.

Although the current implementation is not yet a complete operational railway digital twin, it provides a solid foundation for future work. It demonstrates that railway data can be transformed into SUMO-compatible simulation artefacts, that real-data macro-level simulation is feasible, and that micro-level infrastructure reconstruction can support synthetic railway traffic. Future improvements should focus on route reconstruction, infrastructure consistency, railway-specific operational constraints, and the integration of predictive or real-time data.

Bibliography

[1] The railway infrastructure technical website, 2023.

[2] AGICO Group. What is a railway switch?, 2017.

[3] Michael Behrisch, Laura Bieker, Jakob Erdmann, and Daniel Krajzewicz. Microscopic traffic simulation using sumo. In *2011 IEEE Intelligent Transportation Systems Conference (ITSC)*, pages 1–6, 2011.

[4] Mark M. Dekker, Alexey N. Medvedev, Jan Rombouts, Grzegorz Siudem, and Liubov Tupikina. Modelling railway delay propagation as diffusion-like spreading. *EPJ Data Science*, 11(1):44, 2022.

[5] Eclipse SUMO. Sumo documentation: Calibrator, 2026. Official documentation.

[6] Eclipse SUMO. Sumo documentation: Railways, 2026. Official documentation.

[7] Eclipse SUMO. Sumo user documentation, 2026. Official documentation.

[8] European Union Agency for Railways. Technical specifications for interoperability. European railway interoperability framework.

[9] Lucas Fournier, François Pilo, and Philippe Sanchez. Energy consumption estimation of trains using kinematic modeling and real-time telemetry. *Energies*, 16(4):1720, 2023.

[10] Aidan Fuller, Zhong Fan, Charles Day, and Chris Barlow. Digital twin: Enabling technologies, challenges and open research. *IEEE Access*, 8:108952–108971, 2020.

[11] Ricardo García, María López-García, Luis Cadarso, and Esteve Codina. A mixed-integer linear program for real-time train platforming management. *Transportation Research Procedia*, 62:815–823, 03 2022.

[12] Farhad Ghofrani, Qing He, Rob M. P. Goverde, and Xiang Liu. Progression in railway delay prediction: A review. *IEEE Transactions on Intelligent Transportation Systems*, 19(12):3966–3981, 2018.

[13] Sayantan Ghosh, Ralf Louwerse, and Rob M.P. Goverde. A graph-based approach for station capacity estimation in dense railway networks. *Applied Sciences*, 13(1):1324, 2023.

[14] Rob M. P. Goverde, Francesco Corman, and Andrea D'Ariano. A three-level framework for performance-based railway timetabling. *Transportation Research Part C: Emerging Technologies*, 67:62–83, 2016.

[15] Michael Grieves and John Vickers. Digital twin: Mitigating unpredictable, undesirable emergent behavior in complex systems. In *Transdisciplinary Perspectives on Complex Systems*, pages 85–113. Springer, 2017.

[16] Davide Andrea Guastella, Alejandro Morales-Hernàndez, Bruno Cornelis, and Gianluca Bontempi. Calibration of vehicular traffic simulation models by local optimization. *Transportation*, 2025.

[17] Ingo A. Hansen and Jörn Pachl, editors. *Railway Timetabling and Operations: Analysis, Modelling, Optimisation, Simulation, Performance Evaluation.* Eurailpress, 2014.

[18] International Union of Railways. Uic code 406: Capacity, 2013. Railway capacity methodology.

[19] Atieh Kianinejadoshah, Stefano Ricci, et al. Comparative application of methods for the combined lines-nodes capacity assessment. *International Journal of Transport Development and Integration*, 4(3):243–251, 2020.

[20] Daniel Krajzewicz, Jakob Erdmann, Michael Behrisch, and Laura Bieker-Walz. Recent development and applications of sumo - simulation of urban mobility. *International Journal On Advances in Systems and Measurements*, 3 & 4, 12 2012.

[21] Werner Kritzinger, Matthias Karner, Georg Traar, Jan Henjes, and Wilfried Sihn. Digital twin in manufacturing: A categorical literature review and classification. *IFAC-PapersOnLine*, 51(11):1016–1022, 2018.

[22] Pablo Alvarez Lopez, Michael Behrisch, Laura Bieker-Walz, Jakob Erdmann, Yun-Pang Flötteröd, Robert Hilbrich, Leonhard Lücken, Johannes Rummel, Peter Wagner, and Evamarie Wießner. Microscopic traffic simulation using sumo. In *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*, pages 2575–2582. IEEE, 2018.

[23] Richard M. Lusby, Jesper Larsen, Matthias Ehrgott, and David Ryan. Railway track allocation: Models and methods. *OR Spectrum*, 33:843–883, 2011.

[24] F. Martorell et al. Digital twin technologies for turbomachinery in a life cycle perspective: A review. *Sustainability*, 13(5):2495, 2021.

[25] Taofeeq D. Moshood, James OB. Rotimi, Wajiha Shahzad, and J.A. Bamgbade. Infrastructure digital twin technology: A new paradigm for future construction industry. *Technology in Society*, 77:102519, 2024.

[26] Quang-Dung Nguyen et al. Digital twins for aircraft maintenance and operation: A systematic literature review and an iot-enabled modular architecture. *Journal of Engineering*, 2023.

[27] OpenRail Association. Open source railway designer. Official website and documentation.

[28] OpenTrack Railway Technology. Opentrack railway technology: Railway simulation. Official website.

[29] Jörn Pachl. *Railway Operation and Control.* VTD Rail Publishing, 2018.

[30] Qinglin Qi, Fei Tao, Ying Zuo, and Dongming Zhao. Digital twin service towards smart manufacturing. *Procedia CIRP*, 72:237–242, 2018. 51st CIRP Conference on Manufacturing Systems.

[31] Rail Management Consultants International. Railsys suite. Official website.

[32] Fei Tao, He Zhang, Ang Liu, and Andrew Y. C. Nee. Five-dimension digital twin model and its ten applications. *Computer Integrated Manufacturing Systems*, 25(1):1–18, 2019.

[33] Xin Wang and Stein W Wallace. Stochastic scheduled service network design in the presence of a spot market for excess capacity. *EURO Journal on Transportation and Logistics*, 5(4):393–413, 2016.

[34] Marouane Yassine et al. Digital twin for railway: A comprehensive survey. *ResearchGate*, 2023.

[35] Marouane Yassine et al. A digital twin system for railway infrastructure. *ResearchGate*, 2023.